Dell PowerStore: Clustering and High Availability

April 2026

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White Paper

Abstract

This white paper provides an overview of PowerStore clustering technology along with the highly available features of the platform for the PowerStoreOS releases.

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Executive summary

Overview

This white paper describes the fully redundant hardware and high-availability features that are available on Dell PowerStore. These features are designated to withstand component failures in the system and in the environment, such as network or power failures. If an individual component fails, the storage system can remain online and continue to serve data. The system can also withstand multiple failures if they occur in separate component sets. After the administrator is alerted about the failure, they can order and replace the failed component without any impact.

Audience

This document is intended for IT administrators, storage architects, partners, and Dell Technologies employees. This audience also includes any individuals who evaluate, acquire, manage, operate, or design a Dell networked storage environment using PowerStore systems.

Revisions

Date	Part number/ revision	Description
April 2020	H18157	Initial release: PowerStoreOS 1.0.0
September 2020	H18157.1	Updates to high availability section; other minor updates
April 2021	H18157.2	PowerStoreOS 2.0.0 updates
January 2022	H18157.3	PowerStoreOS 2.1.0 updates; template update
April 2022	H18157.4	PowerStoreOS 2.1.1 updates
July 2022	H18157.5	PowerStoreOS 3.0 updates
August 2022	H18157.6	Minor correction in the *Distributed sparing* section
October 2022	H18157.7	PowerStoreOS 3.2 updates
May 2023	H18157.8	PowerStoreOS 3.5 updates
May 2024	H18157.9	PowerStoreOS 4.0 updates Removed references to PowerStore X
February 2025	H18157.10	PowerStoreOS 4.1 updates
April 2026	H18157.11	PowerStoreOS 4.4 updates • System bond updates • Add appliance updates • File on all appliances

We value your feedback

Dell Technologies and the authors of this document welcome your feedback on this document. Contact the Dell Technologies team by email.

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Note: For links to other documentation for this topic, see the PowerStore Info Hub.

Introduction

Document purpose

Having constant access to data is a critical component in any modern business. If data becomes inaccessible, business operations might be affected and can potentially cause revenue loss. Because of this, IT administrators are tasked with ensuring that every component in the data center has no single point of failure. This white paper details how to cluster PowerStore systems and describes the high availability features that are available. PowerStore is designed for 99.9999% availability (based on the Dell Technologies specification for PowerStore). Actual system availability might vary.

PowerStore product overview

PowerStore achieves new levels of value, flexibility, and simplicity in storage. It uses a container-based microservices architecture, advanced storage technologies, and integrated machine learning to unlock the power of your data. It is a versatile platform with a performance-centric design that delivers scale-up and scale-out capabilities, always-on data reduction, and support for next-generation media.

PowerStore brings the simplicity of the public cloud to on-premises infrastructure, streamlining operations with an integrated machine learning engine and seamless automation. It offers predictive analytics to easily monitor, analyze, and troubleshoot the environment. PowerStore is highly adaptable, providing the flexibility to host specialized workloads directly on the appliance and modernize infrastructure without disruption.

Terminology

The following table provides definitions for some of the terms that are used in this document.

Table 1. Terminology

Term	Definition
Appliance	Solution containing the base enclosure and any attached expansion enclosures.
Base enclosure	Used to reference the enclosure containing both Nodes (Node A and Node B) and 25x NVMe drive slots.
Cluster	One or more appliances in a single grouping and management interface. Clusters are expandable by adding more appliances to the existing cluster, up to the allowed amount for a cluster.
Expansion enclosure	Enclosures that can be attached to a base enclosure to provide additional storage.
Fibre Channel (FC) protocol	A protocol used to perform Internet Protocol (IP) and Small Computer Systems Interface (SCSI) commands over a Fibre Channel network.
File system	A storage resource that can be accessed through file-sharing protocols such as SMB or NFS.
Internet SCSI (iSCSI)	Provides a mechanism for accessing block-level data storage over network connections.
Intra-cluster Management (ICM) Network	An internal management network that provides continuous management connectivity between appliances in the PowerStore cluster.

Term	Definition
Intra-cluster Data (ICD) Network	An internal network that provides continuous storage connectivity between appliances in the PowerStore cluster.
Network-attached storage (NAS) server	A virtualized network-attached storage server that uses the SMB, NFS, FTP, and SFTP protocols to catalog, organize, and transfer files within file system shares and exports. A NAS server, the basis for multitenancy, must be created before you can create file-level storage resources. NAS servers are responsible for the configuration parameters on the set of file systems that it serves.
Network File System (NFS)	An access protocol that allows data access from Linux/UNIX hosts on a network.
Node	Component within an appliance that contains processors and memory. Each appliance consists of two nodes.
NVMe over Fibre Channel (NVMe/FC)	Protocol used to perform Non-Volatile Memory Express (NVMe) commands over a Fibre Channel network.
NVMe over TCP (NVMe/TCP)	Protocol used to perform Non-Volatile Memory Express (NVMe) commands over an Ethernet network.
PowerStore Command Line Interface (PSTCLI)	An interface that allows a user to perform tasks on the storage system by typing commands instead of using the UI.
PowerStore Manager	An HTML5 user interface used to manage PowerStore systems.
PowerStore Q model	Container-based storage system that is running on purpose-built hardware. This storage system supports unified (block and file) workloads or block optimized workloads. The PowerStore Q model supports Quad-Level Cell (QLC) NVMe SSDs for data storage.
PowerStore T model	Container-based storage system that is running on purpose-built hardware. This storage system supports unified (block and file) workloads or block optimized workloads. The PowerStore T model supports Triple-Level Cell (TLC) NVMe SSDs for data storage.
Representational State Transfer (REST) API	A set of resources (objects), operations, and attributes that provide interactive, scripted, and programmatic management control of the PowerStore cluster.
Server Message Block (SMB)	A network file sharing protocol, sometimes referred to as CIFS, used by Microsoft Windows environments. SMB is used to provide access to files and folders from Windows hosts on a network.
Snapshot	A point-in-time view of data stored on a storage resource. A user can recover files from a snapshot, restore a storage resource from a snapshot, or clone the snapshot to provide access to a host.
Storage Policy Based Management (SPBM)	Using policies to control storage-related capabilities for a VM and ensure compliance throughout its life cycle.
Volume	A block-level storage device that can be shared out using a protocol such as iSCSI or Fibre Channel.

Term	Definition
vSphere Virtual Volumes (vVols)	A VMware storage framework that allows VM data to be stored on individual Virtual Volumes. This ability allows for data services to be applied at a VM-level of granularity and according to SPBM. Virtual Volumes can also refer to the individual storage objects that are used to enable this functionality.
vSphere API for Array Integration (VAAI)	A VMware API that improves ESXi host utilization by offloading storage-related tasks to the storage system.
vSphere API for Storage Awareness (VASA)	A VMware vendor-neutral API that enables vSphere to determine the capabilities of a storage system. This feature requires a VASA provider on the storage system for communication.

Clustering

Overview

Every PowerStore appliance is deployed into a PowerStore cluster. There is a minimum of one PowerStore appliance and a maximum of four PowerStore appliances configured in a cluster. The appliances within a cluster can all be a specific model or be a collection of different models. For instance, a cluster may contain multiple appliances of the same model for performance consistency across the cluster or be a mix of models for different use cases. PowerStore T and Q model appliances can also exist within the same cluster. You can also scale down PowerStore clusters by removing appliances from an existing cluster.

Clustering PowerStore appliances can provide many benefits:

- Easy scale-out to increase CPU, memory, storage capacity, and front-end connectivity
- Independent scaling of storage and compute resources
- Centralized management for multi-appliance cluster
- Automated orchestration for host connectivity
- Increased resiliency and fault tolerance

Consider the example shown in the following figure. Users can scale out appliances with different models to create a four-appliance cluster. Besides the scale-out benefit, each appliance can scale up with a different number of expansion enclosures. Finally, each appliance in the cluster can have different media types. For example, one PowerStore model could have QLC NVMe SSD drives while the other three PowerStore models could have TLC NVMe SSD drives. This flexible scale-out and scale-up deployment gives customers the ability to grow their clusters with no dependence on the model number, drive count, or even drive type.

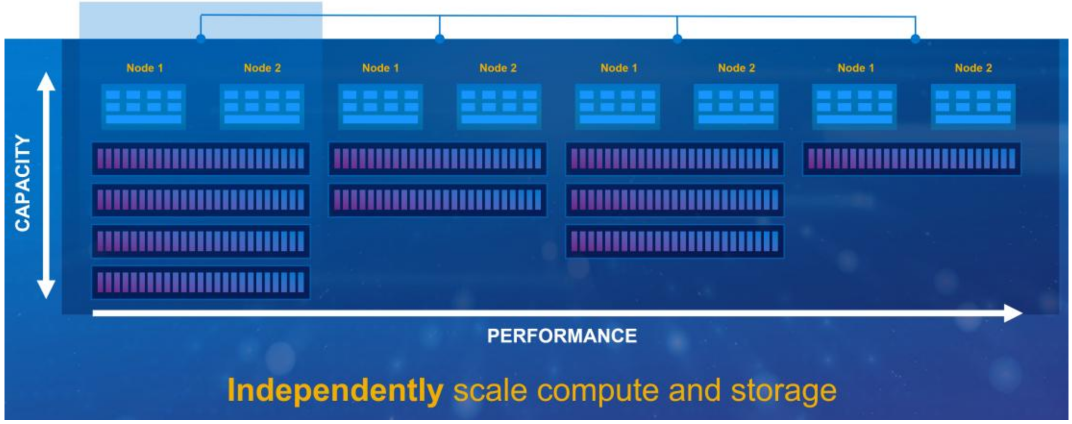

Figure 1. Multi-appliance cluster

Before a PowerStore cluster is created, the minimum cabling requirements must be met. For more information, reference the *PowerStore Quick Start Guide* for the appliance's specific PowerStoreOS version on dell.com/powerstoredocs.

Deployment Modes

PowerStore models support two deployment modes to ensure that the platform delivers maximum performance for each use case. The two deployment modes that are offered are Unified and Block Optimized. Both are detailed in the following sections.

The deployment mode is configured when adding a PowerStore appliance to a cluster. This can occur when creating a new cluster or adding a new appliance to an existing cluster. Once the deployment mode is configured on the appliance, it cannot be changed without a factory reset of the appliance. For this reason, ensure that the proper deployment mode is selected when the system is first configured.

Unified

The default deployment mode is Unified, which is the recommended deployment mode because it provides all capabilities of the PowerStore model appliance. In this configuration, Unified PowerStore model appliances can support block, file, and vVol storage resources. Parts of the underlying hardware resources are reserved for file components and are not available for block resources. For most use cases, the Unified deployment mode is preferred because it supports all resources that the PowerStore model appliance offers.

Block Optimized

The Block Optimized deployment mode for PowerStore models is the alternative deployment mode. Block Optimized systems support block and vVol resources only. All underlying hardware resources are dedicated to block performance in this mode. Block Optimized systems have a higher performance ceiling for block workloads than the same model with a Unified deployment mode.

Single Appliance Clusters

Every PowerStore appliance is deployed into a PowerStore cluster. In PowerStore, each cluster starts with a single appliance. After physically connecting a workstation or laptop to the appliance or discovering the appliance on the network, the user runs the Initial

Configuration Wizard (ICW). The ICW is a GUI driven wizard that assists the user in configuring the appliance into a new cluster.

When completing the Initial Configuration Wizard (ICW), the options provided depend on the code version running on the appliance. In the latest PowerStoreOS versions, the ICW guides the user through:

- Accepting the end user license agreement
- Configuring the cluster information
- Updating the management and service passwords
- Entering the management IP information

Once the wizard is completed, the single appliance PowerStore cluster is created, and storage provisioning can start. Configuring an appliance through PowerStore CLI and REST API is also possible. For more information about the ICW, review the *Dell PowerStore Manager Overview* white paper found on the PowerStore Info Hub.

Initial Configuration Wizard – Cluster Details step

Within the ICW is the Cluster Details step. In this step, the user provides the Cluster Name and the Storage Configuration for the appliance. An example of this step is shown in Figure 2. The Storage Configuration setting configures the appliance as Unified or Block Optimized.

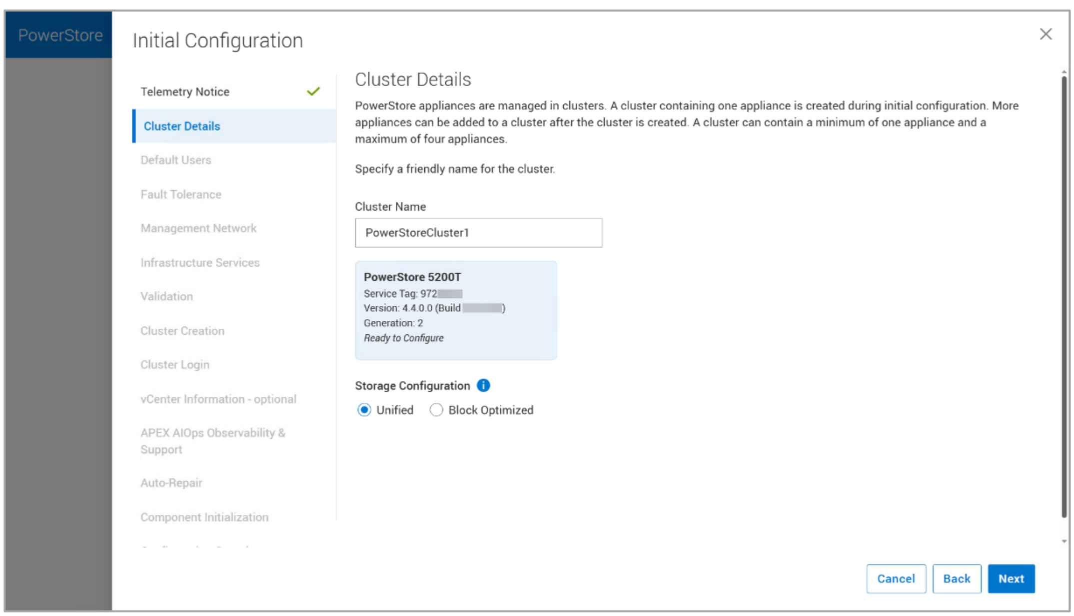

Figure 2. PowerStoreOS 4.4.x and later – Cluster Details step

Unified is the recommended deployment mode as a Unified appliance can support block, file, and vVol storage resources. Block Optimized systems only support block and vVol resources. When selecting between Unified and Block Optimized, consider if the cluster will be expanded with additional appliances in the future. Choosing Unified provides the most flexibility during future cluster expansions. This topic is covered further in the Multi-Appliance Clusters section.

In PowerStoreOS 4.4 and later, only single appliance clusters can be configured using the Initial Configuration Wizard. Any additional appliances must be added after cluster creation. When completing the Initial Configuration Wizard on appliances running PowerStoreOS version 4.3 and earlier, the appliance being deployed scans for other appliances on the network automatically. The **Select Additional Appliances** button is available to add extra appliances to the cluster at this time. The resulting view shows all available appliances that are currently in discovery. The following figure shows an example when the **Select Additional Appliances** option is selected.

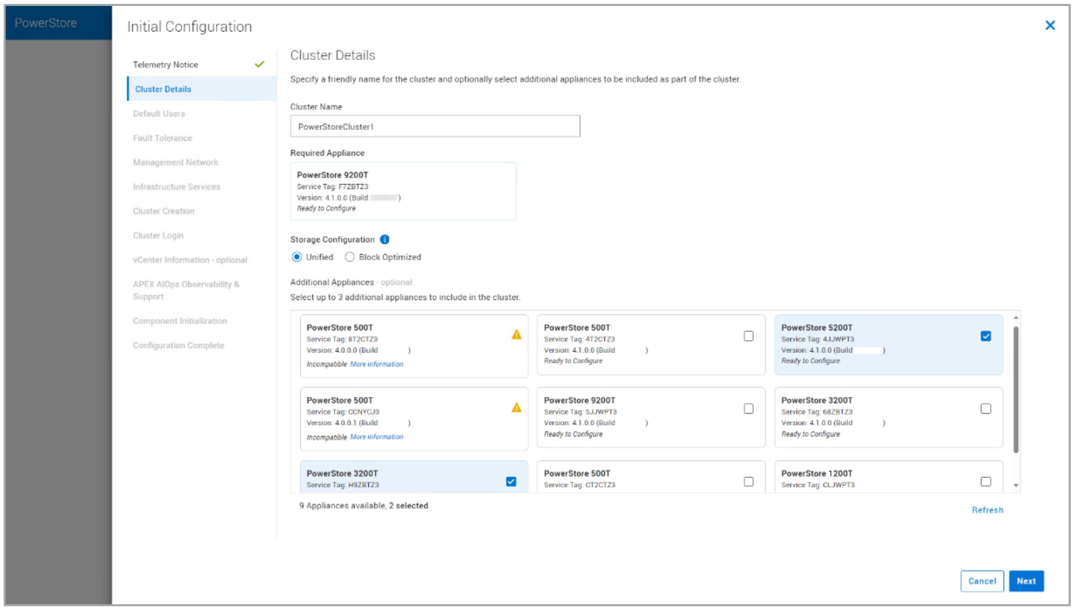

Figure 3. PowerStoreOS 4.3.x and earlier – Cluster Details step

Primary appliance

In PowerStore, each cluster has a primary appliance. The primary appliance is responsible for cluster-level management operations, such as running PowerStore Manager, PowerStore CLI, and REST API services. The primary appliance also coordinates cluster wide activities, such as code upgrades. In PowerStoreOS 4.4 and later, the role of primary appliance is automatically assigned to the appliance being configured regardless of the deployment mode selected.

In PowerStoreOS releases prior to 4.4, multiple appliances could be added to the cluster during cluster creation. The appliance configured in Unified mode would be assigned the role of primary appliance automatically. If all appliances are deployed in Block Optimized mode, the primary appliance is selected by the operating system during cluster creation.

The `svc_cluster_management` service command can be used to view which appliance within the cluster is the primary appliance and to move the role of primary appliance if required. Moving the primary appliance responsibilities is required if the primary appliance is removed from the cluster. Depending on the code version, this may need to be completed manually by the user. This is discussed further in the Remove Appliance section of this paper.

Multi-Appliance Clusters

A PowerStore multi-appliance cluster is a scale-out storage configuration in which two to four Dell PowerStore appliances are joined into a single logical storage system. The cluster operates as one managed entity while distributing storage resources, compute resources, and workloads across multiple appliances to improve performance, capacity, resiliency, and scalability.

All appliances within a cluster can be the same model, which provides performance consistency, or a mix of different appliance models. PowerStore T and Q model appliances can also exist within the same cluster. Appliances can also have different configurations, such as different fault tolerance settings, drive counts, and front-end connectivity options. At any time, a user can expand the cluster by adding an additional appliance or reduce the size of the cluster by removing an appliance. Adding and removing appliances from the cluster are discussed later in this section.

Supported Configurations

When creating a PowerStore cluster, the deployment mode of the individual appliances must be considered. In all PowerStoreOS releases, if the first appliance in the cluster is configured as Block Optimized, all future appliances that are added to the cluster are also configured in Block Optimized mode. This is shown in the **Block Optimized Cluster** table found in Figure 4. If the first appliance in the cluster is in Unified mode, the available deployment mode of the second appliance depends on the PowerStoreOS version running on the cluster being expanded.

PowerStoreOS 4.4 and later

In PowerStoreOS 4.4 and later, a PowerStore cluster supports up to four appliances all deployed in Unified mode. This allows file resources to be created on any appliance within the cluster. When expanding a cluster in the 4.4 release and later, the second appliance added to the cluster sets the deployment mode for all additional appliances. As shown in the **Unified Cluster** table in Figure 4, if the second appliance added to the cluster is configured in Unified mode, all additional appliances will also be added in Unified mode. If the second appliance is added in Block Optimized mode, all additional appliances will be added in Block Optimized mode, as shown in the **Unified and Block Optimized Mixed Cluster** table.

Unified Cluster			
Appliance 1	Appliance 2	Appliance 3	Appliance 4
Unified			
Unified	Unified		
Unified	Unified	Unified	
Unified	Unified	Unified	Unified

Unified and Block Optimized Mixed Cluster			
Appliance 1	Appliance 2	Appliance 3	Appliance 4
Unified			
Unified	Block Optimized		
Unified	Block Optimized	Block Optimized	
Unified	Block Optimized	Block Optimized	Block Optimized

Block Optimized Cluster			
Appliance 1	Appliance 2	Appliance 3	Appliance 4
Block Optimized			
Block Optimized	Block Optimized		
Block Optimized	Block Optimized	Block Optimized	
Block Optimized	Block Optimized	Block Optimized	Block Optimized

Figure 4. Supported cluster configurations

Note: If a single appliance cluster running PowerStoreOS 4.3 or earlier is upgraded to PowerStoreOS 4.4 or later, the PowerStoreOS 4.4 cluster configuration rules apply to cluster expansion.

PowerStoreOS 4.3 and earlier

In PowerStoreOS 4.3 and earlier, Unified mode can only be configured on the first appliance within the cluster. All additional appliances added to the cluster are required to be configured as Block Optimized. Only the Unified and Block Optimized Mixed Cluster and Block Optimized Cluster tables in Figure 4 are applicable to PowerStoreOS version 4.3 and earlier.

Clustering Network

Each node in every appliance that is a part of a PowerStore cluster must communicate with each other. The networks that allow the nodes to communicate are internal to the cluster and are named the Intra-Cluster Management (ICM) and Intra-Cluster Data (ICD) networks. In PowerStore Manager, these networks can be viewed by navigating to **Settings** > **Networking** > **Network IPs**. In single appliance clusters, communication paths internal to the appliance are used.

In multi-appliance PowerStore clusters, the ICM and ICD networks communicate through a link aggregation, also known as a bond. These bonded ports connect to the top-of-rack switch network with native VLAN network packets and use auto-generated IPv6 addresses. All appliances in the cluster should be in the same rack or multiple racks in the same data center. If PowerStore appliances span multiple switches, ensure that the native VLAN is configured on the switch ports and are shared across the switches. Link aggregations are discussed further in the section named Link Aggregation Control Protocol (LACP).

PowerStoreOS 4.4 and later

When deploying a system running PowerStoreOS 4.4 or later, no link aggregations exist on the cluster by default. When adding a second appliance to the cluster, a link aggregation must be specified for the ICM and ICD clustering networks on the existing appliance and the appliance being added to the cluster. The link aggregations used for clustering do not need to be on the same ports on the existing appliance and the appliance being added. The link aggregations used for clustering are not dedicated and can also be used for other purposes such file interfaces, storage networks, replication, and import.

Once a multi-appliance cluster is created, the link aggregations used for clustering are set for the life of the cluster and cannot be modified. If additional appliances are added to the cluster, the existing ICM and ICD interfaces are used for clustering and no additional ports are required. If one or more appliances are removed resulting in a cluster containing a single appliance, the ICM and ICD networks are automatically removed from the link aggregation. If the link aggregation is no longer in use for any purpose, the bond can be deleted.

PowerStoreOS 4.3 and earlier

In PowerStoreOS 4.3 and earlier, a link aggregation is automatically created using ports 0 and 1 of the 4-port card during cluster creation. This link aggregation, also known as the system bond, is automatically assigned the ICM and ICD clustering networks even if clustering of multiple appliances is never implemented. These networks cannot be removed, which also means this bond can never be deleted. In addition to clustering, the link aggregation can also be used for file interfaces, storage networks, replication, and import.

If a single appliance cluster running PowerStoreOS 4.3 and earlier is upgraded to PowerStoreOS 4.4 or later, the ICM and ICD networks are removed from the system bond. The system bond is changed to a user defined link aggregation and can be deleted if no longer in use. If a multi-appliance cluster is upgraded, the ICM and ICD network remain assigned to the ports.

Add Appliance

PowerStore allows scaling up and scaling out a cluster independently. To add more capacity, scale up by adding additional drives and expansion enclosures. To add compute, memory, and connectivity, scale out by adding appliances to an existing cluster. You can add an appliance using the PowerStore Manager, REST API, and PSTCLI.

Consider these general guidelines when adding appliances to an existing cluster:

- The final cluster size cannot exceed four appliances
- You can only add uninitialized appliances
- You can only add one appliance at a time
- The new appliance and the existing cluster must be in a healthy state

- All link aggregations used for the clustering networks must be able to communicate with each on the same native VLAN

To add an appliance to an existing cluster in PowerStore Manager, go to the **Hardware** > **Appliances** page and click the **Add** button. The following sections cover the major steps within the Add Appliance wizard.

Note: The following sections outline the major steps found in PowerStoreOS 4.4 and later. Clusters running earlier codes have slightly different steps due to updates in the 4.4 release.

Cluster Details

This step lists the unconfigured appliances that are available to add to the cluster. This page includes systems that are not faulted and are running a software version within one major release (N-1 or N+1) of the PowerStoreOS running within the cluster. Unconfigured appliances listed as **Incompatible/sync available** will need to synchronize their code to the PowerStoreOS version running on the cluster before the appliance can be added. To do this, select the appliance, click **Next**, then click the "Synchronize" button. When the synchronization is complete, continue with the Add Appliance wizard to add the appliance.

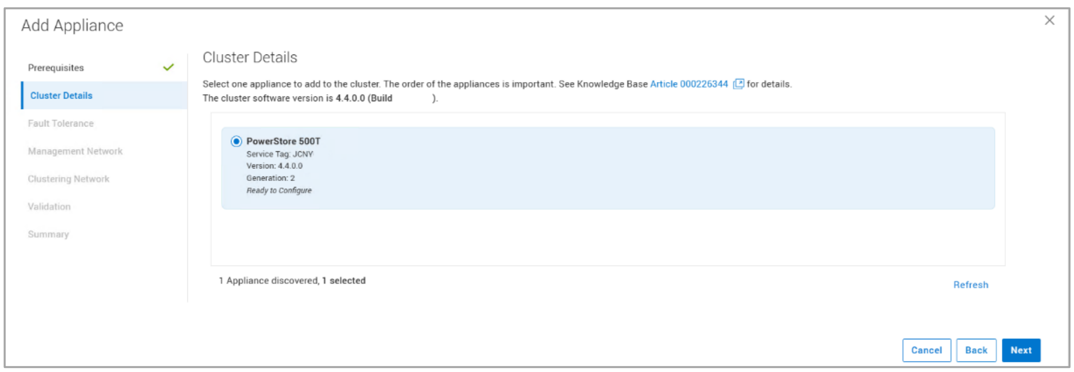

Figure 5. Add Appliance wizard – Cluster Details step

In this example, the cluster includes an appliance in Unified mode, and a second appliance is being added to the cluster. In PowerStoreOS 4.4 and later, the second appliance added to the cluster can be deployed in Unified or Block Optimized mode. As discussed in the Supported Configurations section, the selection at this time determines the appliance mode for any additional appliances added to the cluster in the future.

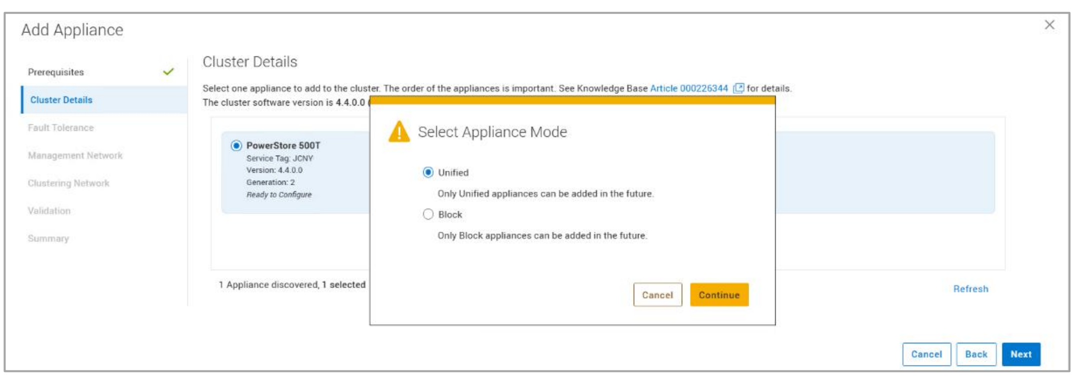

Figure 6. Select Appliance Mode window

If block is selected, the appliance must be rebooted before it can be added to the cluster. This reboot occurs automatically to configure the appliance in block optimized mode. The **Appliance Reboot Required** window is shown below.

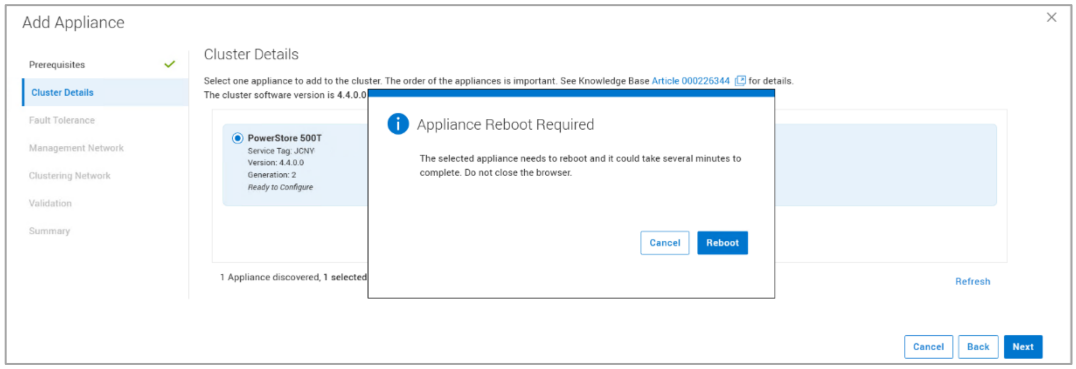

Figure 7. Appliance Reboot Required window

Fault Tolerance

In the Fault Tolerance step, the **Tolerance Level** for the appliance is selected. This setting determines the number of drives that can fail concurrently on an appliance without incurring data loss. The options are **Single Drive Failure** and **Double Drive Failure**. These are discussed further in the Resiliency sets section of this paper.

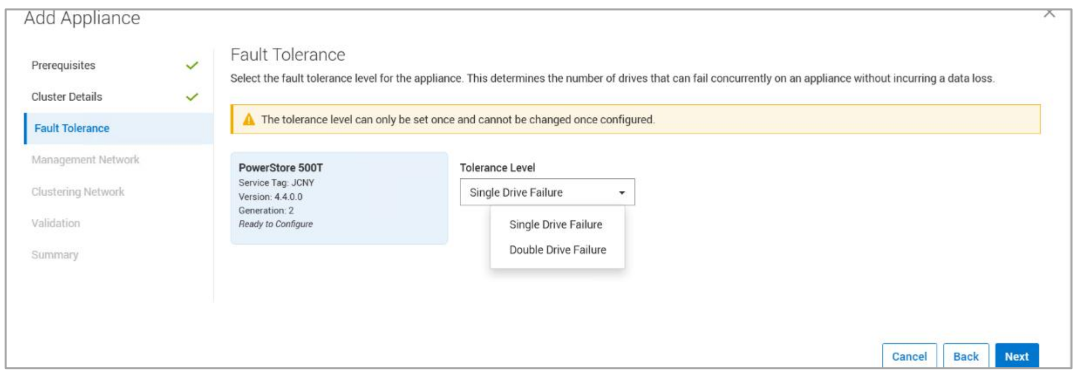

Figure 8. Add Appliance wizard –Fault Tolerance step

Management Network

After selecting an appliance and setting the fault tolerance level, IP addresses for the new appliance must be added if the minimum number of free IPs is not met. There are two methods for adding IP addresses to the cluster, in the Add Appliance workflow or in the Settings page. In the Add Appliance workflow, the user is notified if additional IPs must be added, as shown in Figure 9.

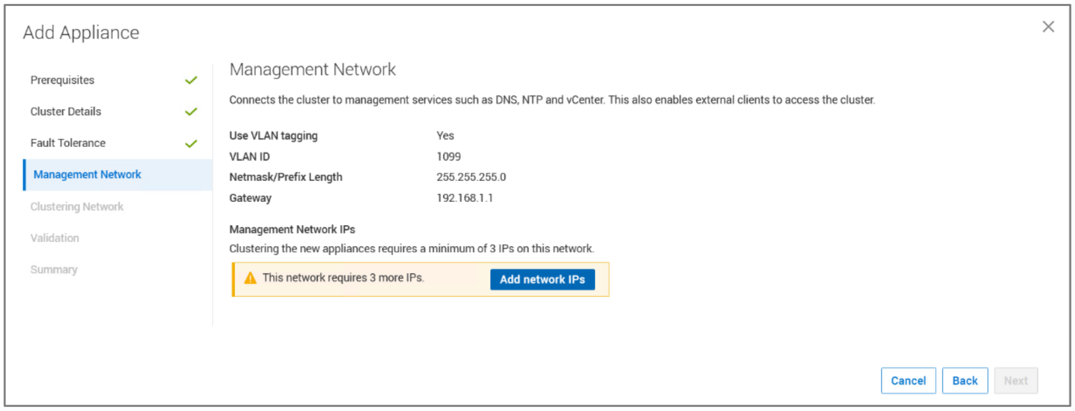

Figure 9. An example of the cluster requiring additional IPs for appliance management

After selecting the **Add network IPs** button, the page in Figure 10 is displayed. For cluster expansion, you must have three or more unused management IP addresses to expand the cluster. To supply additional IP addresses for management, click **Add**. The following figure shows the minimum requirement of three unused IP addresses added in this example.

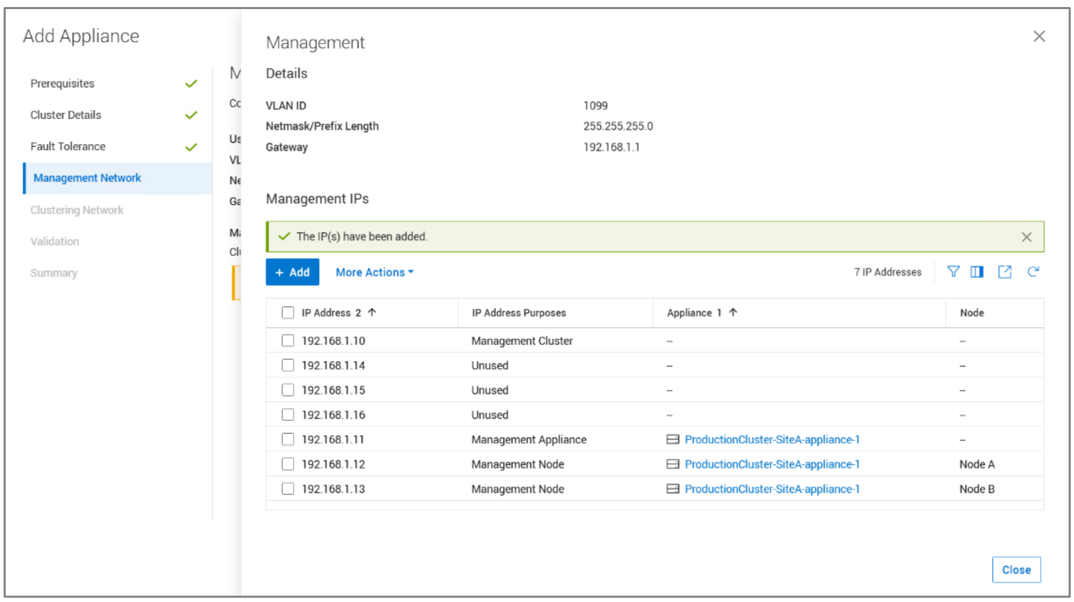

Figure 10. Adding management IPs in the Add Appliance workflow

The alternative method for adding IP addresses is to navigate to **Settings** > **Networking** > **Network IPs** > **Management**. The following figure shows an example of this page. Additional management IPs can be entered in this page, which will be used to manage appliances added to the cluster. This is a way to reserve extras IPs within the cluster for future expansion.

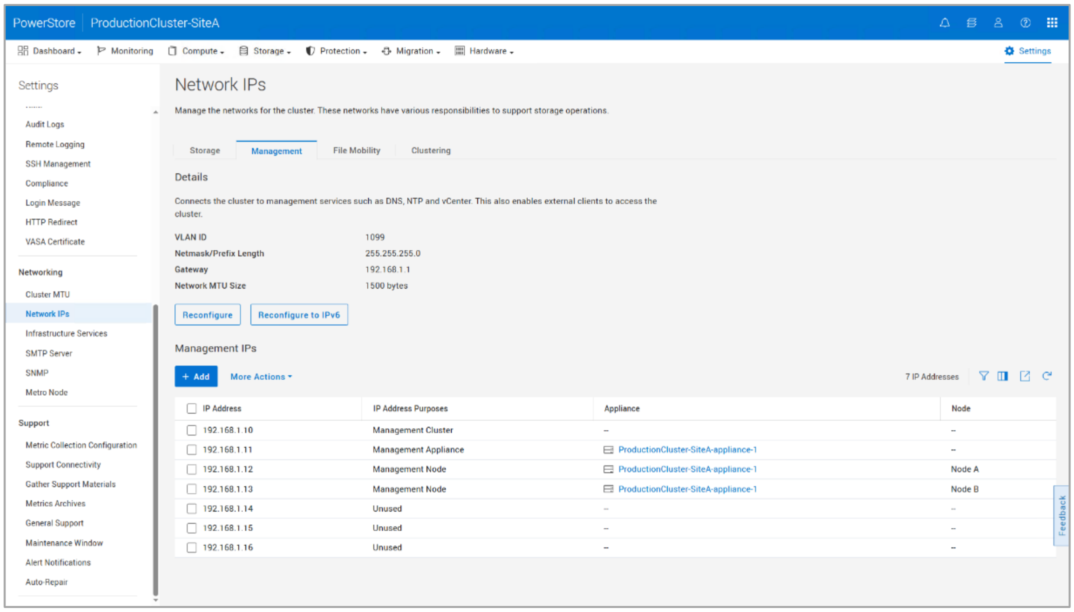

Figure 11. How to add IPs from the Settings page

Clustering Network

The Clustering Network step is new in the PowerStoreOS 4.4 release. This step is used to specify which Link Aggregations the ICD and ICM networks will utilize for cluster communication. In this example of a single appliance cluster, the link aggregation must be specified on the appliance currently within the cluster, and on the appliance being added. If the existing cluster has multiple appliances, ports only need to be specified for the appliance being added. Below is an example of this screen.

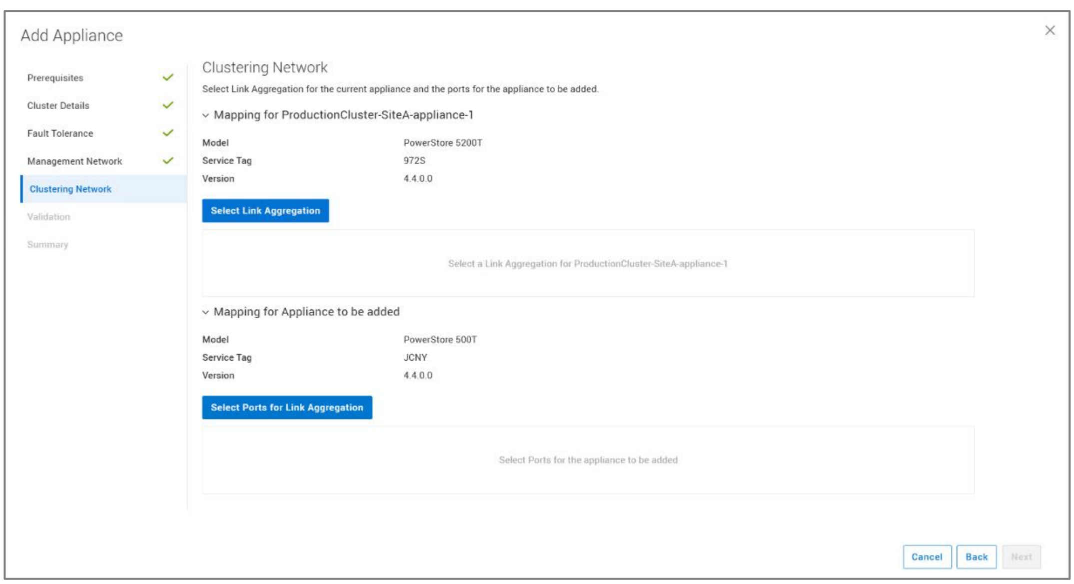

Figure 12. Add appliance – Clustering Network step

Clicking **Select Link Aggregation** opens a window that allows the user to select an existing bond or create a new one for the existing appliance within the cluster. When

Select Ports for Link Aggregation is selected, the window below is displayed. Here, the user can choose which ports will be used for the appliance being added.

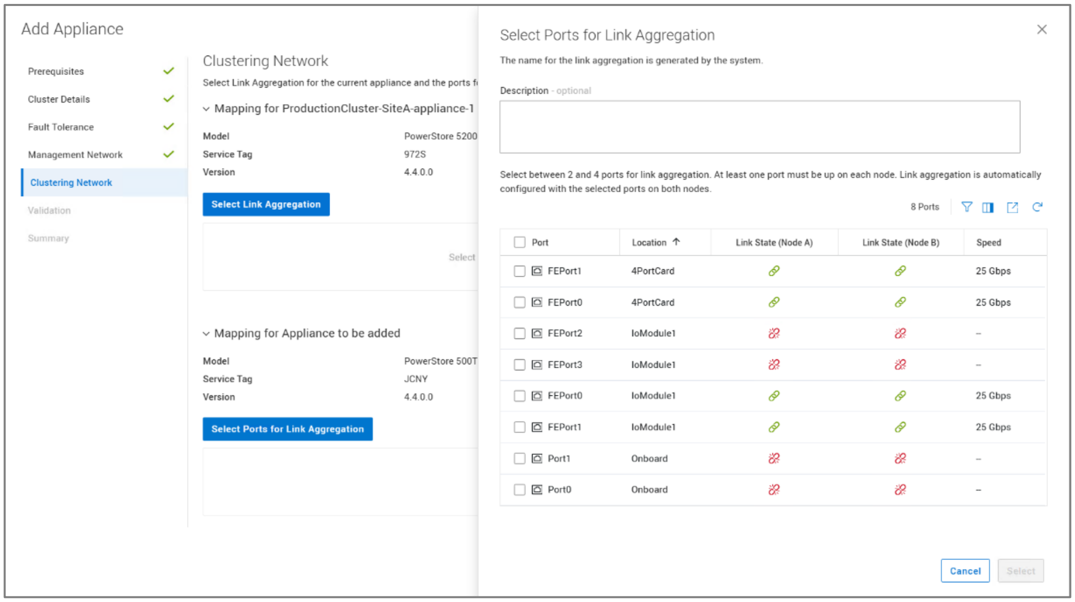

Figure 13. Add appliance – Clustering Network step – Select Ports for Link Aggregation

Once the Link Aggregations are specified, they can be reviewed on the Clustering Network page. An example of the step with information entered can be found below.

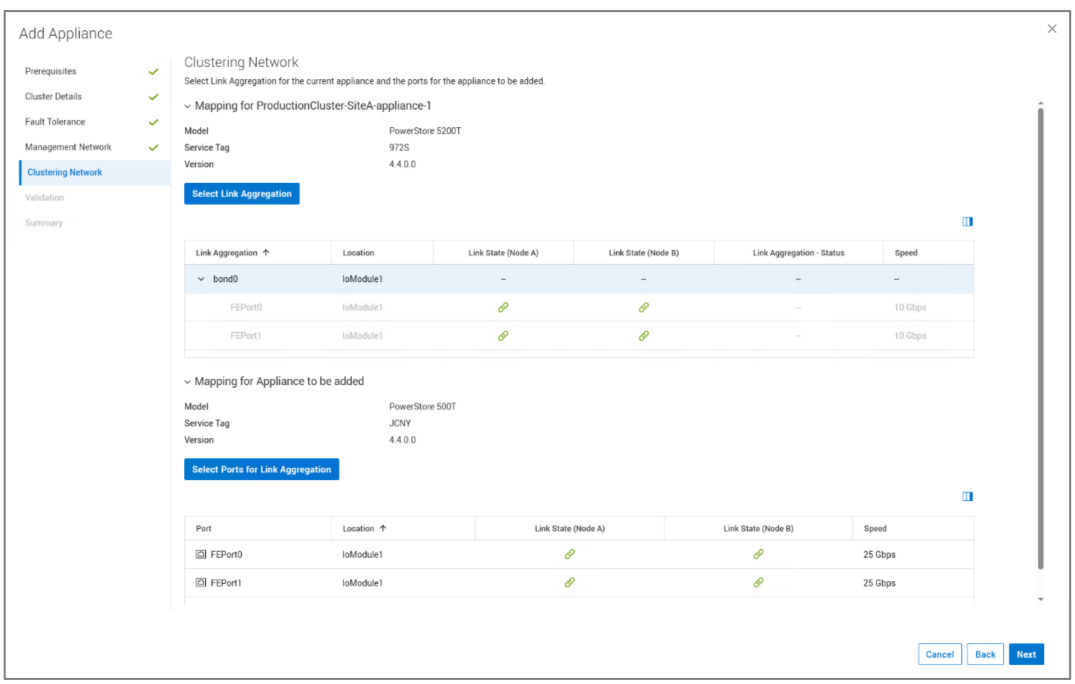

Figure 14. Add appliance – Clustering Network step – Links specified

Validation

In the **Validation** step, a network validation check must be run before completing the **Add Appliance** wizard. Warnings or errors might be displayed after the network validation check is performed. It is recommended to address warnings before beginning a cluster expansion, but any warnings displayed can be bypassed. If there are errors, however, these must be corrected before continuing. If the errors are not addressed, the cluster expansion is blocked from starting. Figure 15 shows an example of the Summary page where the configuration can be validated.

Figure 15. Add appliance – Validation step

Summary

The Summary step of the Add Appliance wizard is shown below. This step is used to confirm the information entered in the wizard. If there are any issues, the **Back** button is available to go back to a previous step. Once the information is confirmed to be correct, click **Add Appliance** to expand the cluster with the selected appliance.

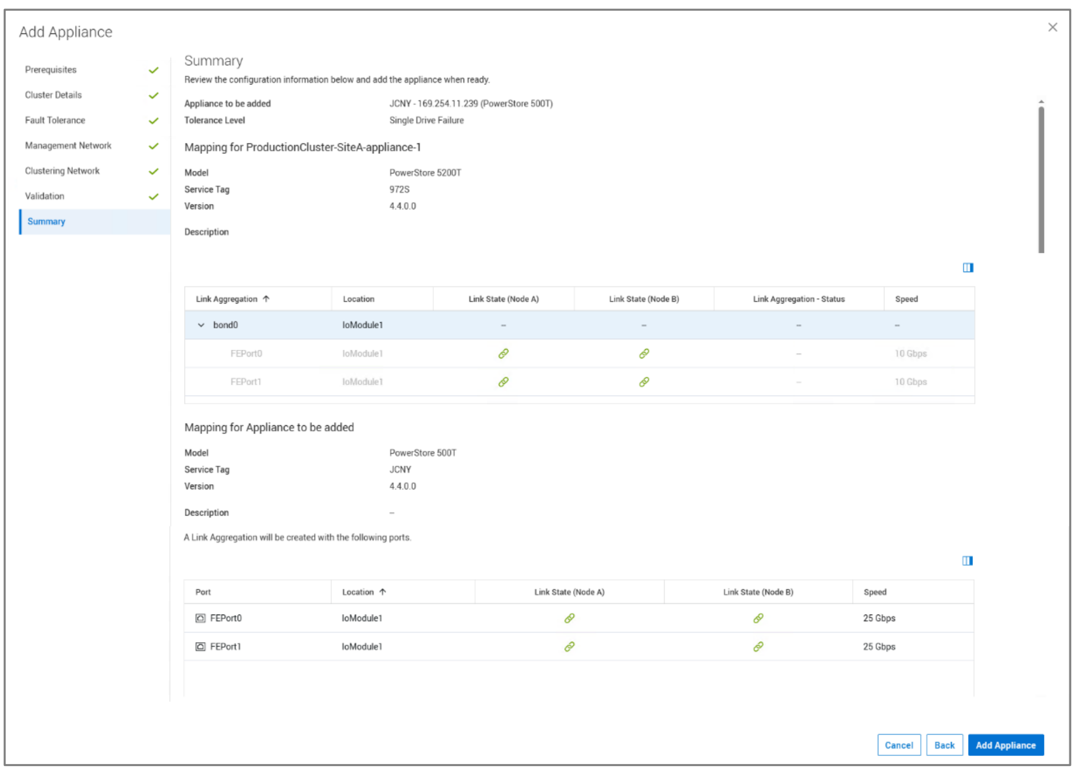

Figure 16. Add appliance – Summary step

Review the job

After the Add Appliance wizard is completed, a new job is created. To see more details about the job, click the **Jobs** icon in the top-right area of PowerStore Manager. Figure 17 shows an example of an Add Appliance job that is running in the background. Other tasks within PowerStore Manager can be completed while the Add Appliance job runs.

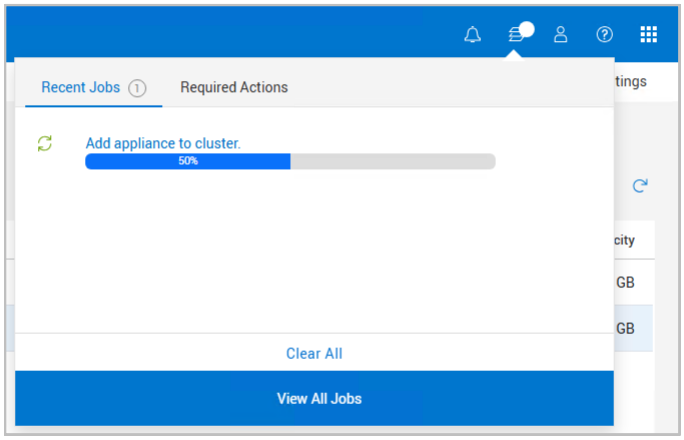

Figure 17. Add appliance job

Remove Appliance

PowerStore clusters can be scaled down by removing individual appliances from the cluster. Removing an appliance from a cluster requires careful planning and consideration. After the appliance is removed from the cluster, the appliance is reimaged and reverted to the original factory settings.

Before starting the appliance removal process, storage objects residing on the appliance need to be removed. Volumes, volume groups, and vVols can be migrated to other appliances in the cluster. This process can be performed in PowerStore Manager by navigating to **Hardware** > **Appliances**. Once on the **Appliances** page select the appliance to be removed and select **More Actions** > **Migrate**. The wizard-based workflow helps to move storage resources to other appliances in the cluster.

Table 2 shows the high-level steps for removing a PowerStore appliance from the cluster. This information is also outlined in detail within PowerStore Online Help. Search for **Remove an appliance from a cluster** within Online Help for more information and specific guidance. Figure 18 shows the information provided to the user after selecting the appliance and clicking **Remove**. The hyperlink navigates the user to the help page mentioned previously.

Note: If a cluster includes one appliance in Unified mode and one or more appliances in Block Optimized mode, removing the Unified appliance forces the cluster to become Block Optimized. No additional appliances can be added in Unified mode if the Unified appliance is removed. If all Block Optimized appliances are removed, appliances in Unified Mode can be added to the cluster.

Table 2. High-level steps to remove an appliance from a cluster

	Operation
1	Move the Primary Appliance operations off the appliance manually if required
2	Disable support notifications on the cluster
3	Disable resource balancer on the appliance being removed
4	Migrate any remaining resources to other appliances in the cluster
5	Remove the appliance from the cluster

1. Move the Primary Appliance operations off the appliance manually if required

In a PowerStore cluster, one appliance has the role of primary appliance. One task of the primary appliance is to run the PowerStore Manager service. Before the primary appliance can be removed from a cluster, the role of primary must be moved to another appliance within the cluster. In PowerStoreOS 4.4 and later, the remove appliance process automatically migrates the primary role to another PowerStore in the cluster if required. While not required, the primary role may be moved prior to starting the remove appliance process. In releases prior to PowerStoreOS 4.4, if a user attempts to remove the primary appliance from the cluster, the job fails, and a failure message is provided. On these releases the primary role must be migrated manually.

One method to determine the primary appliance is to enable SSH management on the cluster and use an SSH client to connect to the cluster IP. The service tag displayed within the command prompt identifies the primary appliance. The `svc_cluster_management` service command can also be used to identify the primary appliance. This command is also used to move the primary role to another appliance within the cluster if required. This process is outlined in the Online Help page referenced earlier in this section.

2. Disable support notifications on the cluster

When removing an appliance, it is suggested disable support notifications on the cluster. This prevents call home alerts from occurring during the maintenance activity. In PowerStore Manager navigate to **Settings** > **Support** > **Maintenance Window**. The appliance removal process typically takes less than one hour, but it is suggested to set a longer duration in case unforeseen issues arise. The maintenance window should be disabled once the removal process completes.

3. Disable resource balancer on the appliance being removed

Resource balancer is the cluster process that identifies which appliance a storage resource should be assigned to. This is primarily used when creating new storage resources, and the placement of the resource is set to auto. Disabling resource balancer on the appliance that is being removed ensures no additional resources are assigned to the appliance automatically. The `svc_appliance_provisioning` service command is used to disable this setting. This process is outlined in the Online Help page referenced earlier in this section.

4. Migrate any remaining resources to other appliances in the cluster

At this time any resources remaining on the appliance being removed need to be migrated to another appliance within the cluster or deleted. To migrate resources or confirm none are present on the appliance, select the checkbox in front of the system on the Appliance page, then select **More Actions** > **Migrate**. If no resources remain on the appliance, the window displays **No objects found**.

5. Remove the appliance from the cluster

Removing an appliance from a cluster can be completed in PowerStore Manager or using the `svc_remove_appliance` service script. In PowerStore Manager, select the checkbox in front of the system on the Appliances page, then select Remove. Clicking Remove within the Remove Appliance window will create a background job and remove the appliance from the cluster. An example of the Remove Appliance window can be seen in Figure 18.

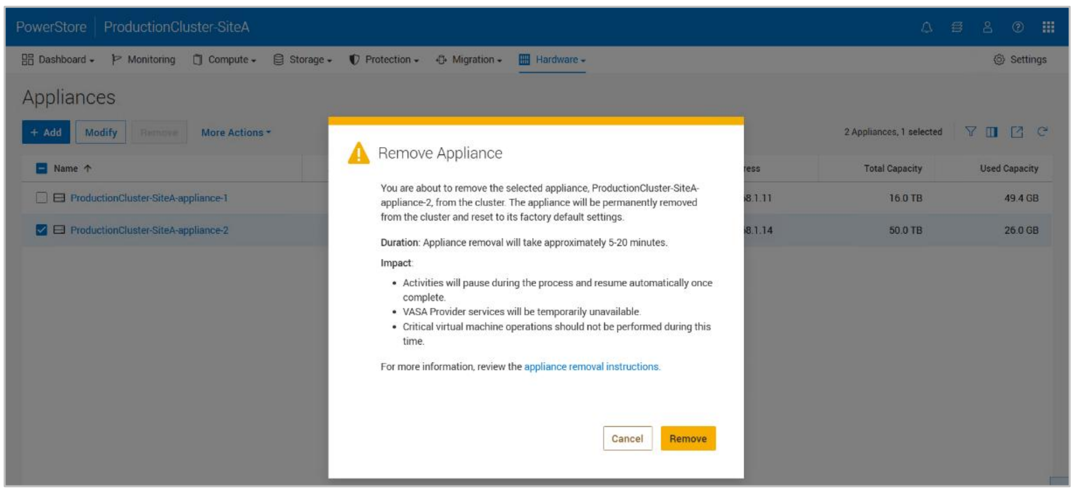

Figure 18. Remove appliance message

VMware integration

PowerStore offers deep integration with VMware vSphere. These integrations include VAAI and VASA support, event notifications, snapshot management, storage containers for virtual volumes (vVols), and virtual machine discovery and monitoring in PowerStore Manager. By default, when a cluster is created, a single storage container is automatically created. Storage containers are used to present vVol storage from PowerStore to vSphere. vSphere mounts the storage container as a vVol datastore and makes it available for VM storage. This vVol datastore is then accessible by external ESXi hosts. If an administrator has a multi-appliance cluster, the total capacity of a storage container spans the aggregation of all appliances in the cluster. For more information about VMware Integration with PowerStore, see the white paper *Dell PowerStore: Virtualization Integration*.

Resource balancer

In the modern data center, administrators must be quick and agile to support mission-critical applications. PowerStore includes an intelligent analytical engine that is built into the PowerStore operating system. This engine offers many benefits such as helping administrators make decisions that are based on the initial placement of data, and automatically balancing node performance within an appliance. It also assists with resource migrations or storage expansions that are based on analytics and capacity forecasting.

Initial placement of data

When storage administrators create storage resources, the resource balancer can provide many benefits. For example, when a multi-appliance cluster is deployed, the resource balancer intelligently and automatically places newly created volumes or NAS servers on different appliances in the cluster. This placement is based on the deployment mode of the appliances, and which appliance has the most unused capacity. If you choose to override the decision of automatic placement, you can manually select the appliance on which to place the resource.

The following figure shows an example of a volume with the **Placement** setting of **Auto**. PowerStore will choose which appliance to assign the resource automatically. The drop-down menu allows the user to specify another appliance if required. Note that when volumes or volume groups are created, they are only placed on a single appliance and do not span multiple appliances in the cluster. After these storage resources are created, they do not move until they are migrated manually to another appliance in the cluster. The migration of storage resources within a cluster is discussed in the following sections of this paper.

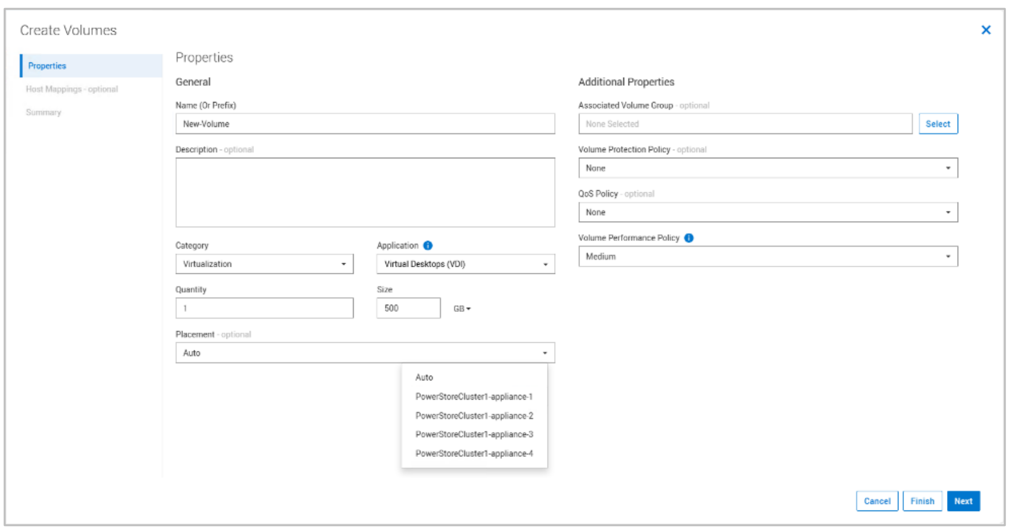

Figure 19. Automatic placement of volumes or manual assignment

Volume groups

In PowerStore, when creating a volume group, all corresponding members in the volume group are placed on the same appliance. If creating a volume group using existing volumes, the volumes could reside on different appliances in the cluster. In this situation, the storage resources must be manually migrated to a single appliance before placing them into a volume group.

NAS Servers

When creating an NAS server, the appliance selection is set to Auto. When set to Auto, the resource balancer automatically selects which appliance to create the NAS server if multiple appliances within the cluster are deployed in Unified mode. A user can override the appliance placement setting of Auto and choose which appliance to create the NAS server if desired. All file systems created on the NAS server will be created on the same appliance as the NAS server.

Capacity forecasting

PowerStore provides details on when a cluster might run out of space. These details help when planning to scale a cluster based on business needs. To best inform these decisions, PowerStore maintains historical statistics about the consumed storage on an appliance to intelligently predict how this storage will be used over time.

To provide the most accurate information, the system collects statistics over 15 days. After 15 days, PowerStore Manager displays a forecast of when a system might run out of space. The following figure shows an example of viewing capacity forecasting in PowerStore Manager.

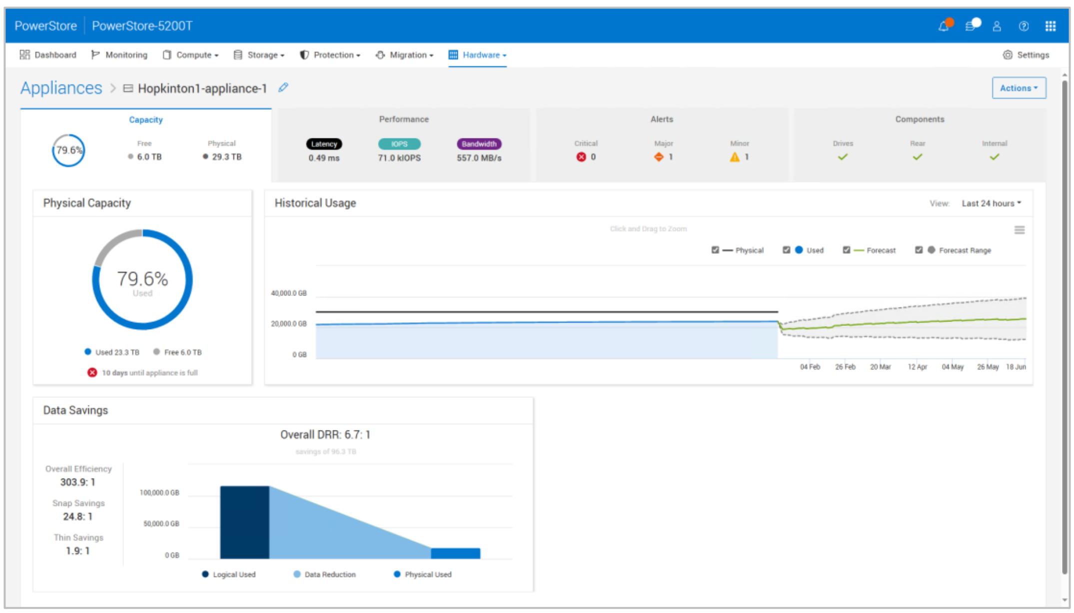

Figure 20. Capacity forecasting

Dynamic Node Affinity

When mapping volumes to hosts, PowerStore dynamically sets a node for preferred host access, also known as Node Affinity, as shown in the following figure. As more storage resources are created and used over time, there might be situations where one PowerStore node is more heavily used than the peer node. In this scenario, PowerStore may automatically change the Node Affinity of one or more storage resources if certain performance metrics checks are met.

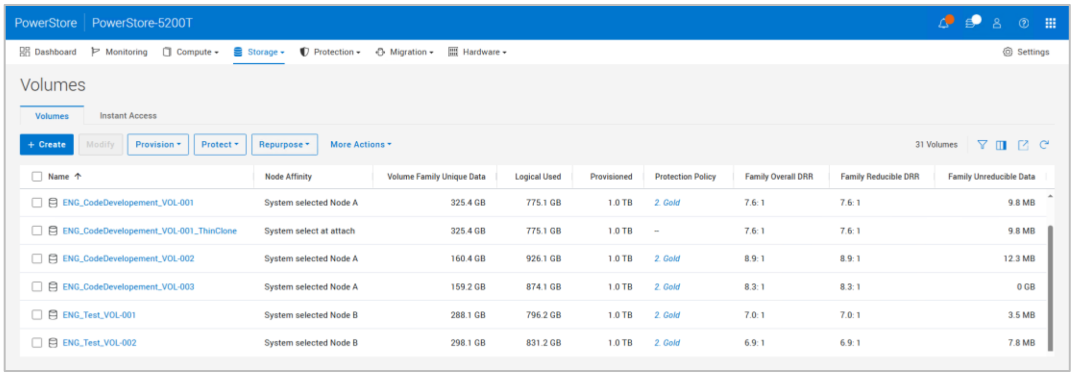

Figure 21. Node Affinity

The first performance check compares the read and write IOPs of the nodes. The CPU utilization for each node is checked to ensure that at least one is exceeding 50%. The second check is to see if the system is exceeding a latency check as reported by the data path engine. PowerStore checks these performance metrics every 30 minutes. If all three of these performance checks are met throughout the duration of the 30 minutes, PowerStore dynamically changes the node affinity of the volume and the active/optimized path to the host. This change is transparent and non-disruptive to the host. PowerStoreOS 4.0 and later expands dynamic node affinity to support vVols as well. It leverages the same resource balancer engine that is used for volumes and provides the same functionality and benefits.

Migration overview

After a volume or volume group is created, capacity and demand may change over time. PowerStore supports moving a volume, volume group, virtual machine, or vVol to another appliance within the cluster. This operation can be performed with a manual or PowerStore assisted migration using PowerStore Manager, REST API, and PSTCLI. These methods are detailed in the next sections of this paper. For more information regarding vVol migrations, see the document *Dell PowerStore: Virtualization Integration*.

Before the start of a migration job, verify the following:

- Ensure that the host has connectivity to the source and destination appliances with the appropriate protocol
- Ensure that the host object in PowerStore Manager shows initiators going to the source and destination appliances
- From the host, perform a rescan of the storage object that will be migrated

After verifying the above steps, a migration job can be started that is non-disruptive and transparent to the host. Although it is not visible to the end user, the storage resource and its data are transferred to the new appliance within the cluster by using asynchronous replication technology. When the storage resource is moved to the destination, all associated storage objects, such as snapshots and clones, are also moved to the destination. After the data transfer is complete, the host switches paths automatically to the destination appliance and the migration job is complete.

The following figure shows an example of a multi-appliance cluster that is made up of four appliances. This example involves moving a storage resource from Appliance 1 to

Appliance 3. In this example the host does not have connectivity to Appliance 2 or Appliance 4. This means that the storage resource would not be able to migrate to those appliances until host connectivity is established.

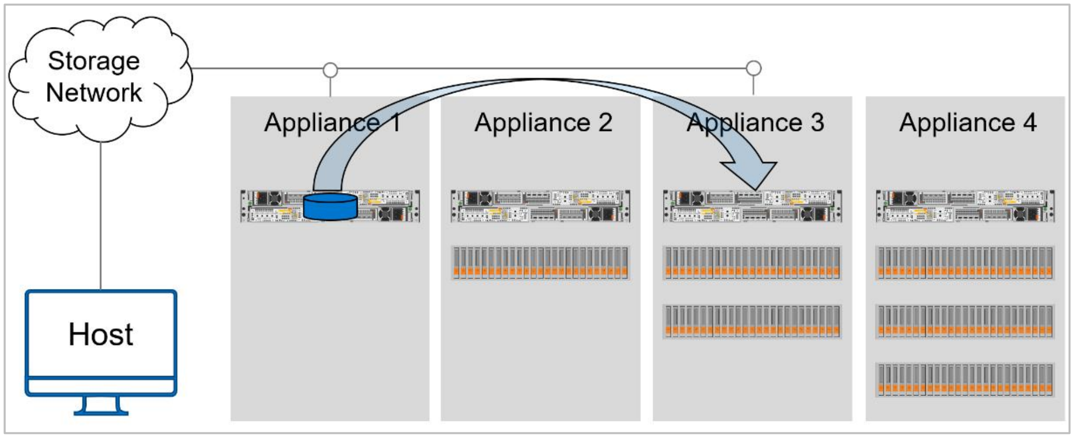

Figure 22. Volume migration example

Manual migration

A manual migration can be used to move storage resources such as a volume, volume group, virtual machine, or vVol to another appliance within the cluster. Before moving the storage resource, ensure that paths are seen from the PowerStore Manager and that a host rescan has been performed. After the host initiator paths are verified, perform a manual migration on the storage resource. If the host does not have access to the destination appliance, the migration will fail.

Performing a manual migration in PowerStore for a volume, volume group, virtual machine, or vVol depends on the storage resource:

- Volume: Select **Storage** > **Volumes** to view a list of all available volumes in the cluster. Select a volume to migrate and click **More Actions** > **Migrate.**
- Volume group: Select **Storage** > **Volume Groups** to view a list of all available volumes in the cluster. Select a volume group to migrate and click **More Actions** > **Migrate.**
- Virtual Machine: Select **Compute** > **Virtual Machine** to view a list of virtual machines in the cluster. Select the virtual machine to move, then click **More Actions** > **Migrate**.
- vVol: Select **Storage** > **Storage Containers** and select the storage container where the virtual machine is deployed. Select the **Virtual Volumes** tab to view all associated vVols in the cluster, select a vVol to migrate, and click **Migrate**. Although individual vVol migration is possible, it is recommended to keep all vVols of the same VM on a single appliance to minimize the fault domain and improve availability. It is suggested to migrate VMs instead to ensure vVol co-location.

The following figure depicts a volume that is created on a multi-appliance cluster that is being migrated. The volume resides on appliance 1 and the wizard displays a list of the other appliances in the cluster. To migrate the volume, select the desired destination appliance and click **Next**. This action runs multiple pre-checks and creates the data migration session. Before the migration can proceed, a host rescan is required to ensure host connectivity is established to the new resource on the new appliance. The steps presented on the Summary step must be followed to complete the migration process.

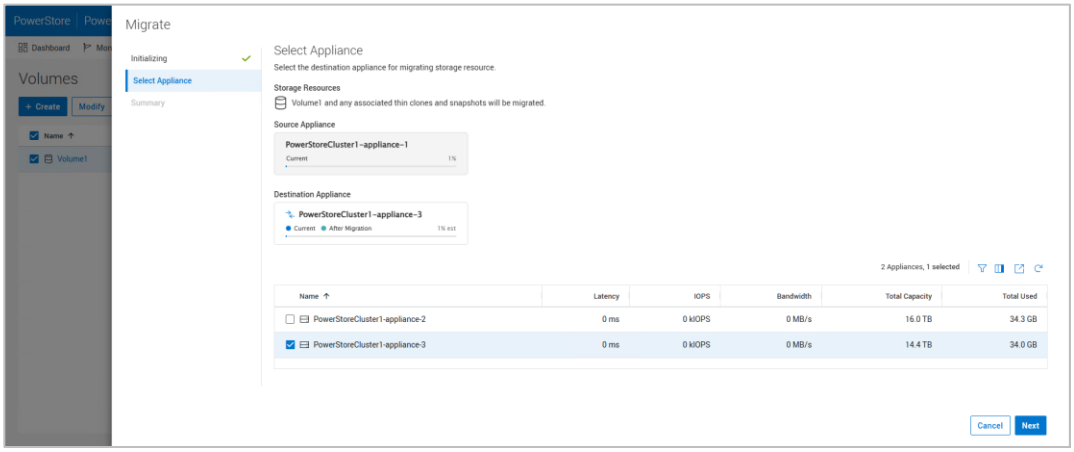

Figure 23. Start volume migration

Manual migrations for volumes, volume groups, virtual volumes, virtual machines, and VMware Site Recovery Manager replication groups can also be started from the **Hardware** > **Appliances** page. This method allows a user to easily view the resources created on a particular appliance and migrate some or all of them to other appliances within the cluster. To start a migration on the **Appliances** page, select an appliance, click **More Actions** > **Migrate**. An example of the Migration workflow is shown in Figure 24.

As shown in the figure below, the user has the options of **All Available Objects** or **Select Object Manually**. The **All Available Objects** option is typically used when the appliance will be powered off and the resources must remain online, or the appliance is being removed from the cluster and all objects must be removed. When **Select Objects Manually** is used, the user can select specific resources to move. Based on the selections, the **Non-Migratable Storage Objects** step in the workflow informs the user of objects that currently cannot be migrated. Possible reasons are space constraints, object type, and appliance/host connectivity. Objects that are part of an active import or migration session are not eligible for migration.

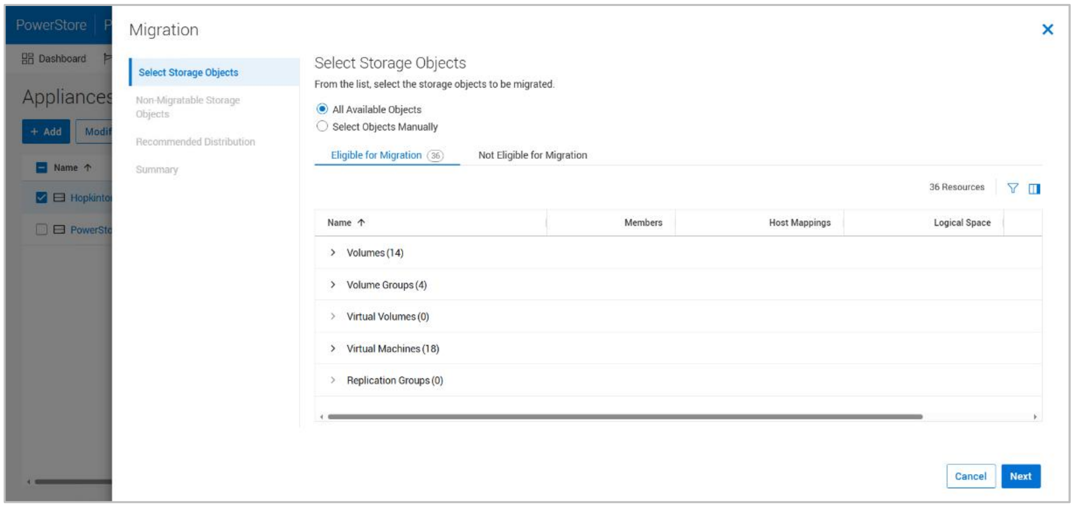

Figure 24. Appliance Migration workflow

On the **Recommended Distribution** step, PowerStore displays which appliance is the best destination for the storage resources. The selection can be manually overridden by selecting the **Modify Recommended Distribution** option. The pencil icon in the **Destination Appliance** column can be used to manually select a different appliance within the cluster than the one suggested. An example of this step with the **Modify Recommended Distribution** option expanded can be seen in the figure below.

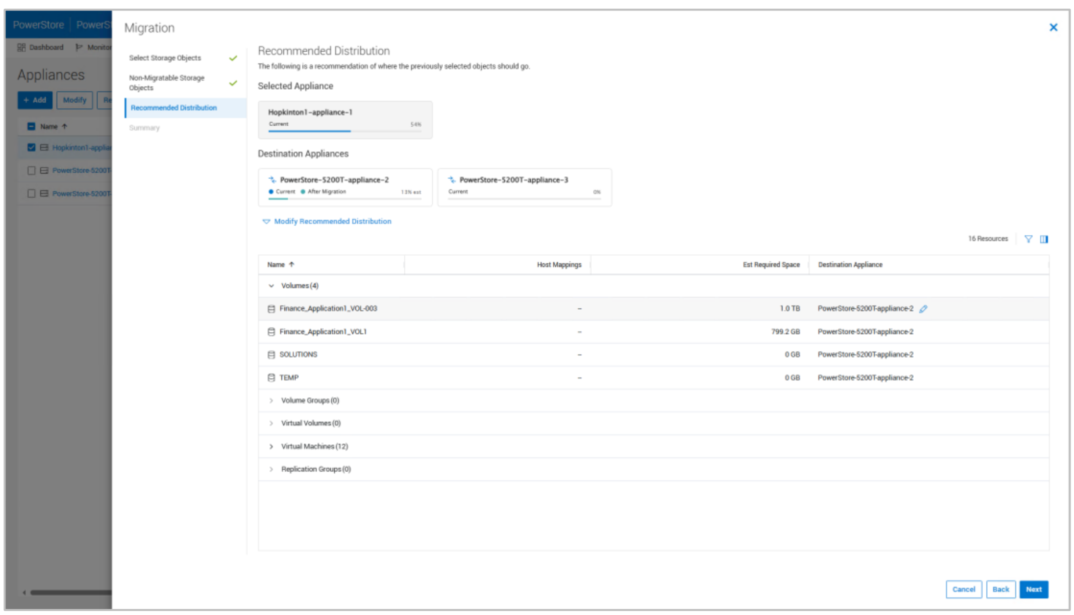

Figure 25. Appliance Migration workflow – Recommended Distribution step

At the end of the wizard, PowerStore provides a list of associated hosts that require a rescan. After the rescan is completed, PowerStore automatically generates the migration sessions and starts the migration jobs.

Assisted migration

PowerStore periodically monitors storage resource utilization across all appliances within the cluster. If an appliance is approaching the maximum usable capacity, the system generates migration recommendations as shown in the following figure. These recommendations are generated based on factors such as drive wear, appliance capacity, and health. An administrator can view assisted-migration recommendations from the capacity alerts generated on the cluster. If the administrator accepts a migration recommendation, a migration session is automatically created. Any snapshots or clones that are associated with the original storage resources are migrated over to the new appliance, as with a manual migration.

The figure below is an example of a capacity alert for an appliance that is projected to run out of space in 10 days. After selecting the alert, under **Next Steps** is a link to **Assisted Migration**. Clicking the link displays the screen in the figure below. Here, PowerStore provides recommendations for internal cluster migrations of resources. The **View Recommended Distribution** option can be selected to view more information on PowerStore's suggested remediation steps. Clicking **Next** accepts the recommendations and proceeds with the migrations.

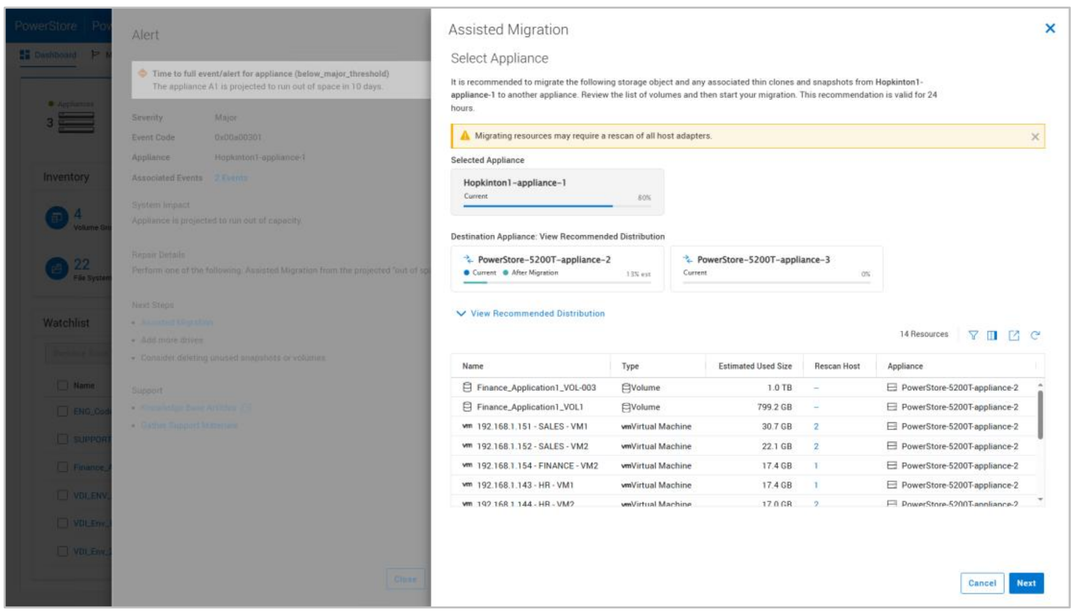

Figure 26. Assisted migration recommendations

Migration states

Migration sessions can be viewed on the **Migration** > **Internal Migrations** > **Migrations** page. Administrators can pause, resume, or delete any active migration sessions through PowerStore Manager, REST API, or PSTCLI. The following table shows a list of all the possible states for a migration session.

Table 3. Migration Session States

State	Definition
Initializing	Migration session starts and stays in this state until the session initialization completes
Initialized	Migration session transitions to this state when the session initialization completes
Synchronizing	Background data copy is in progress

State	Definition
Idle	Migration session transitions to this state when the initial background data copy completes
Cutting_Over	A final portion of the data copy is performed in this state, and the ownership of the storage resource is transferred to the new appliance.
Deleting	Migration session is being deleted
Completed	Migration session is completed, and it is safe to delete the session
Pausing	Migration session transitions to this state when the pause command is issued
Paused	Migration session is Paused, and user intervention is required to resume the session
System_Paused	Migration session transitions to this state if it encounters any error, and user may resume or delete the migration after resolving the error
Resuming	Migration session background copy is resumed
Failed	Migration session encountered an error

High availability

Introduction

Although PowerStore appliances support multi-appliance configurations to increase redundancy and fault tolerance throughout the cluster, each PowerStore appliance also has two nodes that make up a High Availability (HA) pair. PowerStore features fully redundant hardware and includes several high-availability features. These features are designed to withstand component failures within the system and in the environment. Examples of these failures include network or power issues. If an individual component fails, the storage system can remain online and continue to serve data. The system can also withstand multiple failures if they occur in separate component sets. After a failure alert is issued, the failed component can be ordered and replaced without impact. This section reviews PowerStore redundancy and fault tolerance within the platform and the cluster.

Hardware redundancy

PowerStore has a dual-node architecture which includes two identical nodes for redundancy. It features an active/active controller configuration in which both nodes service I/O simultaneously. This feature increases hardware efficiency since there are no requirements for idle-standby hardware. The base enclosure includes both the nodes and the drives. The number of drives and drive types depends on the PowerStore appliance model. For details about the components of a PowerStore appliance, see the appropriate *Hardware Information Guide* on dell.com/powerstoredocs. See also the white paper *Dell PowerStore: Introduction to the Platform* on the PowerStore Info Hub.

Management software

The management software on the primary appliance in the cluster runs PowerStore Manager, the management interface (cluster IP), and other services such as REST API and PSTCLI. The management resources are balanced between both nodes of the appliance to promote better CPU utilization across the nodes. PowerStore Manager runs on the secondary node while the other management services run on the primary node. Figure 27 shows an example of a primary node from the hardware properties page, which is highlighted by going to **Hardware** > **Appliances** > **[Appliance]** > **Components** > **Rear View**. Under **BaseEnclosure** on the left, the primary node is identified by having **(primary)** next to the node name.

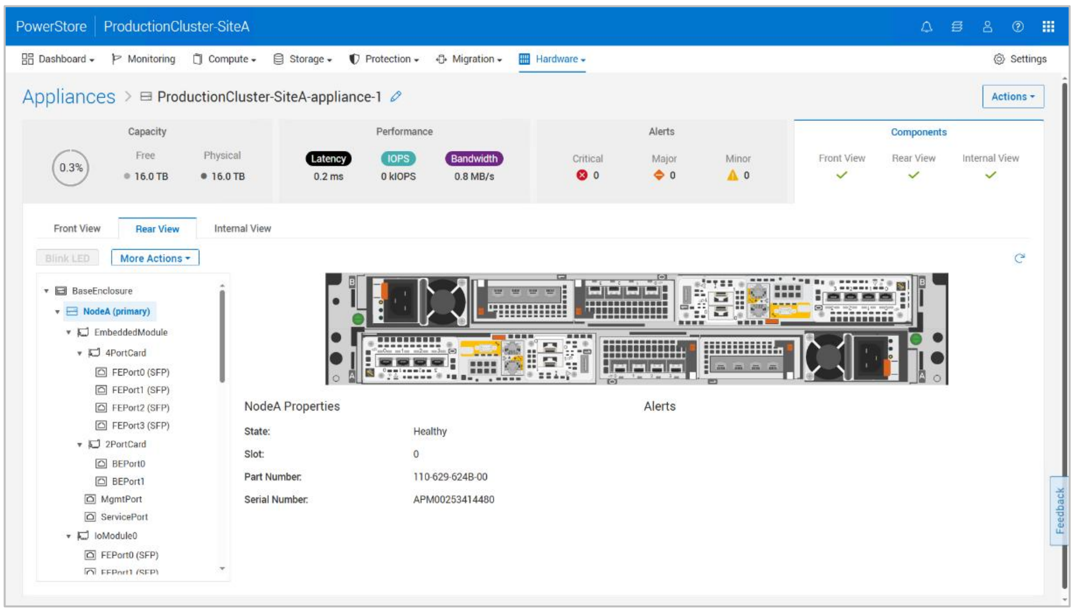

Figure 27. Primary node

If a node reboots, or the management connection goes down, the services running on that node automatically fail over to the peer node. After a failover, it might take several minutes for all services to start completely as resource I/O takes precedence. Users that are logged in to PowerStore Manager during the failover may see a message indicating that the connection has been lost. When the failover process completes, access to PowerStore Manager is automatically restored, however users can also manually refresh the browser to regain access. Host access to storage resources is prioritized and available before PowerStore Manager is accessible.

After the failover, the services temporarily run on the node. When the peer node has recovered from the failover event, the services automatically rebalance after five minutes. During the rebalance, PowerStore Manager might lose connection while services are brought back up on the recovered node. Note that after failover events, the primary node might change to the peer node and will remain this way until the system is rebooted or failed over again.

Link Aggregation Control Protocol (LACP)

Link loss can be caused by many environmental factors, such as cable or switch port failure. It is suggested to configure high availability on the ports to protect against these types of failure scenarios.

Link aggregation combines multiple network connections into one logical link. This provides increased throughput by distributing traffic across multiple connections and provides redundancy in case one connection fails. If connection loss is detected, the link is immediately disabled, and traffic is automatically moved to the surviving links in the aggregation to avoid disruption. The switch should be properly configured to add the ports back to the aggregation when the connection is restored. Although link aggregations provide more overall bandwidth, each individual client still runs through a single port. PowerStore systems use the Link Aggregation Control Protocol (LACP) IEEE 802.3ad standard.

In PowerStoreOS 3.0 and later, additional link aggregations, also called user defined bonds, can be configured with two to four ports on the appliance. These ports can come from the same I/O Module, from different I/O Modules, or across I/O Modules and Ethernet ports on the embedded module. When creating a link aggregation, all ports within the aggregation should be cabled on both nodes and have the same speed, duplex settings, and MTU size. If a failover were to occur, the peer node uses the same ports.

It is suggested to configure LACP on the switches for user-created bonded ports. If cabling bonded ports across switches, it is recommended to configure Virtual Link Trunking interconnect (VLTi) with LACP or equivalent technology on the switch to take advantage of the bond's active/active capabilities. Link aggregations are created from the **Hardware** > **Ports** page in PowerStore Manager (Figure 28).

Note: In releases prior to PowerStoreOS 4.1, creating a Link Aggregation was completed from the **Hardware** > **[appliance]** > **Ports** tab.

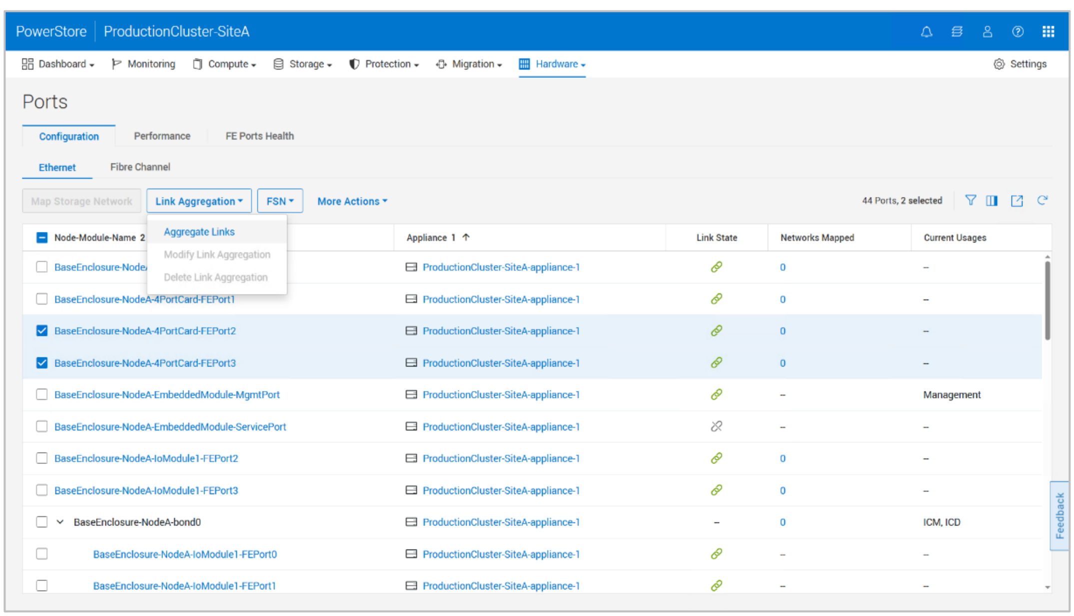

Figure 28. Create Link Aggregation

A single link aggregation can be used for PowerStore ports connecting to the same switch or different switches. For switches that are stacked, link aggregation provides multiple paths to multiple switches for redundancy purposes. Figure 29 shows a cabling diagram for an LACP configuration on 4-port card ports 0 and 1. Each port on node A connects to a different Dell switch, and the configuration is mirrored on node B. For switches that are stacked, VLT, or equivalent technology for a different switch vendor must be configured. A link aggregation group is then configured on the blue paths in the example below, and separately for the green paths.

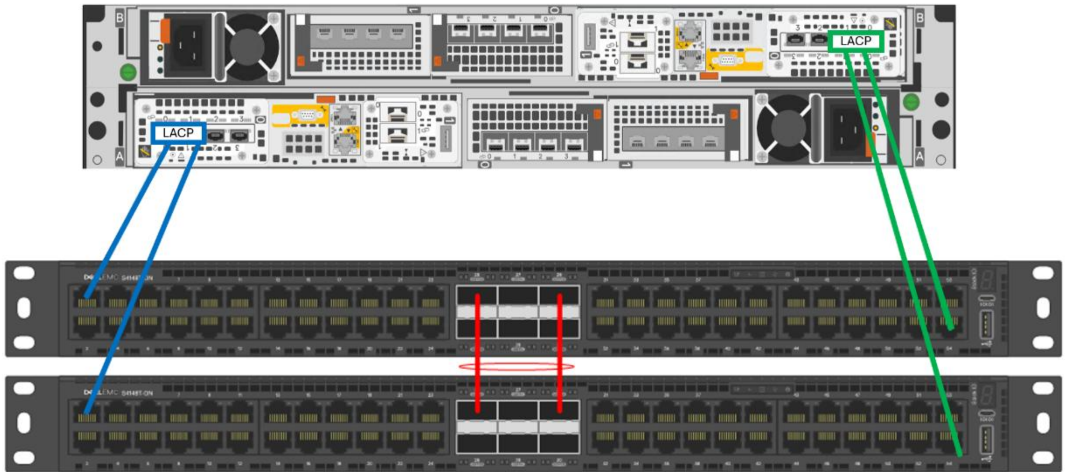

Figure 29. Link aggregation example 1

For networks that contain switches that are not interconnected, a single link aggregation can be used to provide redundancy to one network switch, while a second aggregation can be used for redundancy to another. In Figure 30, the first link aggregation is created on ports 0 and 1 on the 4-port card, shown in blue, and connects to the top switch. A second link aggregation is created on ports 2 and 3 of the 4-port card, shown in green, and connects to the bottom switch. A link aggregation group then needs to be configured for each of the four groups of cables. On node A, a link aggregation group would be configured on the switch for the blue cables to the top switch, and separately on the green paths to the bottom switch. This is then repeated for the ports on node B.

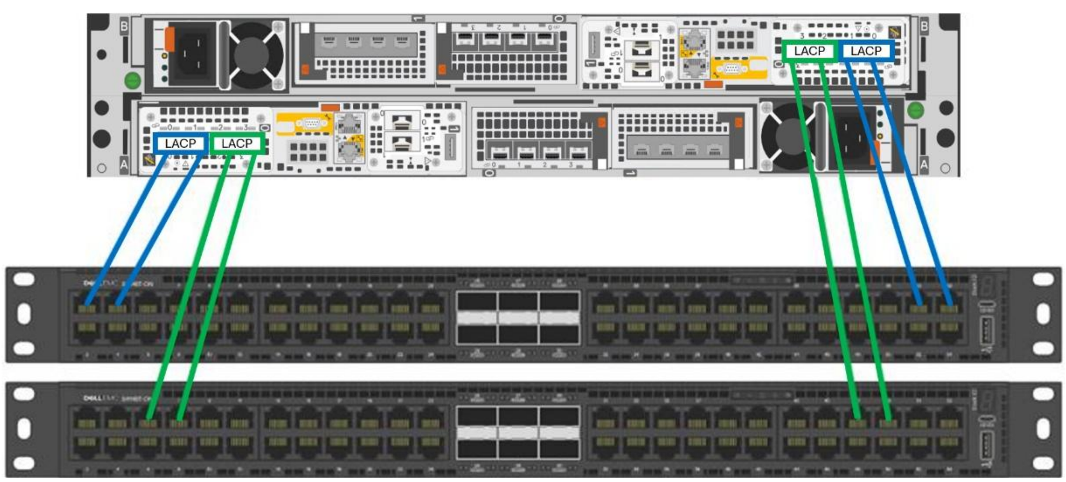

Figure 30. Link aggregation example 2

For client access, NAS Servers include one or more network interfaces that are created on aggregated links for redundancy. Starting in PowerStoreOS 4.0, users can map storage (iSCSI) and replication networks to user defined link aggregated ports. Previously, storage and replication networks were limited to the system bond (bond0) (PowerStoreOS 4.3 and earlier) or any individual ethernet port capable of supporting a storage network. PowerStore bonds also support enabling multiple uses at the same time. File, storage (iSCSI), replication, and import traffic can all share the same bond if required.

Note: User defined link aggregations do not support the NVMe/TCP protocol. However, NVMe/TCP can be utilized on the system created bond (bond0) in PowerStoreOS 4.3 and earlier. If an appliance running PowerStoreOS 4.3 and earlier is upgraded to PowerStoreOS 4.4 and later, NVMe/TCP support remains for the lifetime of bond0. Once bond0 is deleted, NVMe/TCP support is no longer possible on link aggregations.

When replicating from one system to another, it is recommended to configure link aggregations the same way on both systems. If a link aggregation with the same name is not found on the destination system, the interfaces on the destination NAS Server are created without a port assignment and the user must manually assign a port. Alternatively, if a nonmatching configuration is required, users can override the interfaces on the destination NAS Server and assign them to a valid port. Otherwise, data access becomes unavailable in the event of a failover.

Link aggregation should also be configured at the host level/client level to provide resiliency against port or cable failures. Depending on the vendor, this might also be referred to as trunking, bonding, or NIC teaming. See the vendor’s documentation for more information.

Fail-Safe Networking (FSN)

PowerStoreOS 3.5 adds Fail-Safe Networking (FSN) support for file interfaces. FSN is a high-availability feature that allows configuring ports in a primary/backup configuration. Under normal circumstances, the primary ports are designated as active and are used to service I/O. If all primary ports of an FSN go offline, the backup ports automatically become active and continue to service I/O. This enables redundancy in case of port, cable, or switch failure. When the primary ports are restored, the system automatically makes the primary ports active again.

FSN can be leveraged to increase resiliency in file networks, especially when Multi-Chassis Link Aggregation (MC-LAG) is not configured on the top-of-rack switches. MC-LAG enables the ability to create link aggregations across multiple switches. Without MC-LAG, link aggregations are limited to a single physical switch. If that switch goes offline, access to the NAS servers on that node becomes unavailable. Configuring FSN across multiple physical switches enables data access to continue even if a switch goes offline.

FSN is designed to be transparent to the switch, which enables it to work with any switch vendor and does not require any switch configuration. The FSN exposes a single MAC address to the network, so it appears as a single link. An FSN can potentially have multiple IP addresses on it if it is used for multiple NAS server interfaces.

An FSN can consist of individual ports, Link Aggregations, or a combination of the two. When used in conjunction with LA, multiple ports can be used as part of the active or backup part of the FSN. Leveraging both FSN and LA together provides high availability and load balancing. If the primary side of the FSN uses a LA, all ports of the LA must go down for the backup side to become active. Note that in PowerStoreOS 4.3 and earlier you cannot put the system bond into an FSN. In PowerStoreOS 4.4 and later, a link aggregation created on ports 0 and 1 on the 4-port card can be added to an FSN.

An FSN can be created using a different configuration on the primary compared to the backup side, if desired. Members of an FSN can have different speed/duplex settings but must have the same MTU. Ports from different I/O modules can be used in the FSN. FSNs can also be created with more ports in the active side compared to the backup ports. Note that having a mismatched configuration may have performance implications in failure scenarios.

For more information about configuring FSN, see the white paper *Dell PowerStore: File Capabilities* on the Dell Technologies Info Hub.

Dynamic Resiliency Engine (DRE)

Enterprise-class storage systems require high levels of reliability and protection from data loss and latent drive failures. Traditional data protection schemes are based on RAID groups of a fixed layout that protect a volume's data. The bandwidth and rebuild speed in this traditional design are limited by the number of drives participating in the group. Also, the speed of the rebuild is limited by the number of tolerated drive failures of that RAID level. For example, a 6+2 could achieve the read speed of six drives but only sustain two drives worth of rebuild speed in the case of a dual-drive failure.

The reliability of the data being protected depends heavily on the Bit Error Rate (BER) of the drive, the amount of data that must be rebuilt, and the number of drives. As capacity and drive counts increase, it becomes more difficult to maintain reliability when traditional RAID protection schemes are used.

Furthermore, economics and reliability are key buying decisions. The cost of a storage solution is driven by the cost of the drives and ultimately, it is price/performance and effective capacity ($/IOP and $/Effective Capacity). As storage and drive capacity grows, the following occurs:

- Performance scales
- Relative cost of the controller diminishes
- The probability of encountering drive failures increases
- Protection and system metadata overhead increases, impacting effective storage capacity
- Higher rebuild speeds are needed to maintain reliability

DRE overview

The PowerStore Dynamic Resiliency Engine (DRE) is a 100% software-based approach to redundancy that is more distributed, automated, and efficient than traditional RAID. It meets RAID 6 and/or RAID 5 parity requirements with superior resiliency and at a lower cost. PowerStore implements proprietary algorithms where every drive is partitioned into multiple virtual chunks and redundancy extents are created by using the chunks across several drives.

It automatically consumes the drives within an appliance and creates appropriate redundancy using all the drives. This process improves overall performance and allows performance to scale as more drives are added to the appliance. Data written to a volume can be spread across any number of drives within an appliance. As new drives are added, the data is automatically rebalanced.

In PowerStoreOS 4.0, a change to DRE allows the system to free previously reserved space which helps increase the usable capacity of the system. Once an appliance is upgraded from a previous release to 4.0 or later, the space will automatically be available as usable capacity within the system. The amount of space being freed directly depends on the appliance model and drive configuration.

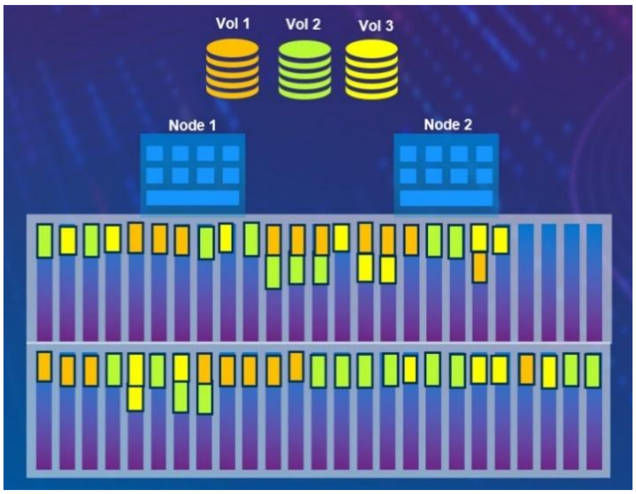

Figure 31. Data placement in PowerStore

Distributed sparing

Unlike traditional RAID protection strategies, PowerStore does not require dedicated spare drives. Spare space is distributed across the entire appliance, and a small chunk of space is reserved from each drive used for sparing if a drive fails. A single drive’s worth of spare space is reserved for every resiliency set in an appliance. Resiliency sets are explained later in this section.

When a drive fails, only the portion of the drive which has data written will be rebuilt. By doing so, the spare capacity is efficiently managed by consuming only the required space. This feature also shortens rebuild time because only data that has been written to the drive must be rebuilt.

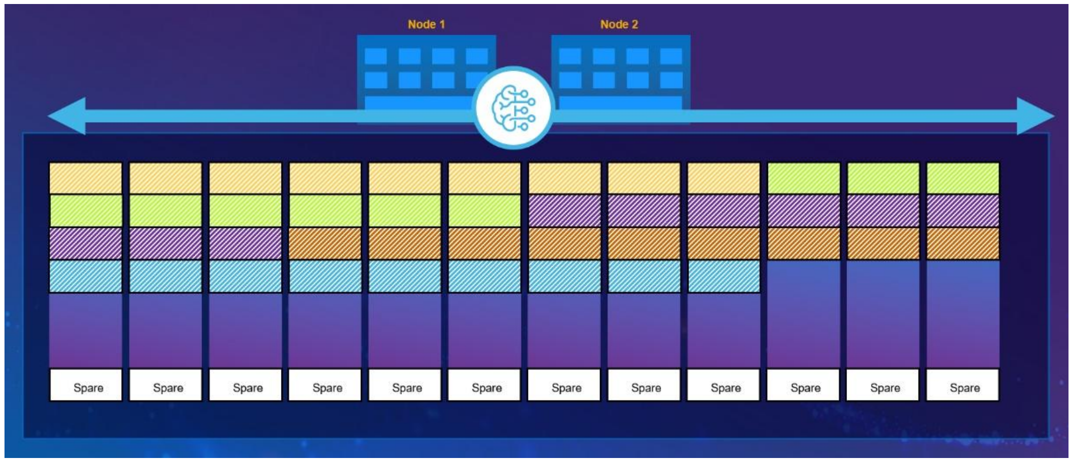

Figure 32. Distributed sparing

Resiliency sets

PowerStore implements resiliency sets to improve reliability while minimizing spare overhead. Having multiple failure domains (resiliency sets) increases the reliability of the system since it allows the appliance to tolerate a drive failure in each of the sets if the failure occurs simultaneously.

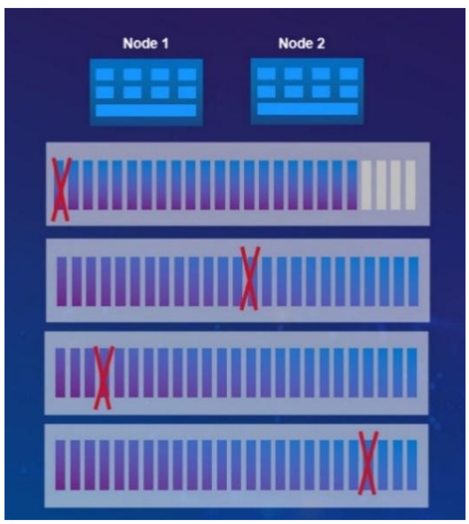

Figure 33. Tolerance for simultaneous drive failure in multiple resiliency sets

An appliance's fault tolerance level is set during cluster creation or when an appliance is added to an existing cluster. The fault tolerance level sets the number of drives that can fail concurrently within a resiliency set without incurring data loss, which can be configured as Single Drive Failure or Double Drive Failure. Single Drive Failure configures 25-drive resiliency sets, while Double Drive Failure configures 50-drive resiliency sets. If deploying a multi-appliance cluster, you can mix different appliances with different drive failure tolerance levels in the same cluster, as shown in the following figure.

Note: PowerStore Q models require double-drive failure tolerance.

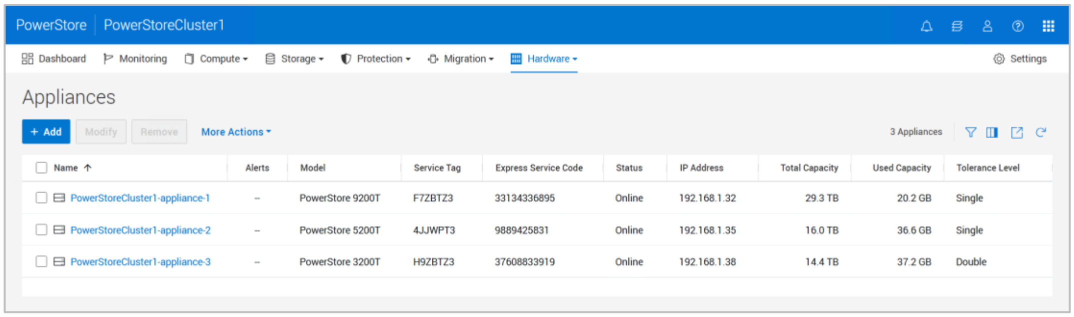

Figure 34. Appliance fault tolerance level

As capacity and drives are added to the system over time, resiliency sets dynamically increase and expand. For example, you might have an appliance that has a single-drive-fault-tolerance set. This means that the system is configured with a 25-drive resiliency set. If you add a 26th drive to the system, the resiliency set dynamically splits into two sets. Furthermore, resiliency sets can span across physical enclosures based on the number of drives in the appliance and have a mix of drive sizes.

Resiliency sets offer the following key benefits:

- Enterprise class availability
- Faster rebuild times with distributed sparing
  - Rebuild smaller chunks of the drive simultaneously to multiple drives in the appliance

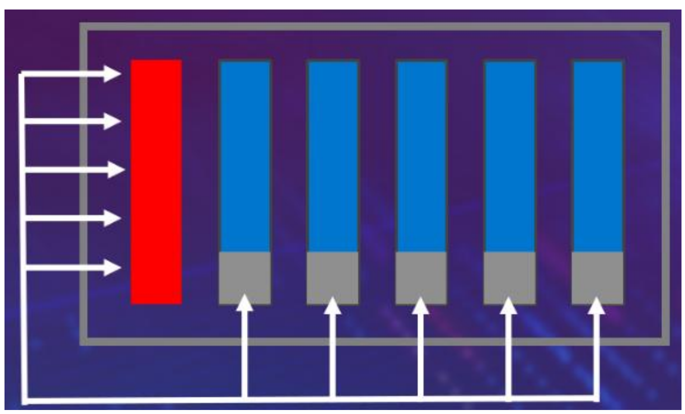

Figure 35. Parallel rebuild of single drive to distributed spare space

- Intelligent infrastructure:
  - Automatically allocate unused user space to replenish spare space to handle multiple failures. DRE dynamically transfers unused user capacity to replenish spare capacity if there is sufficient unused capacity available on the appliance.
  - Intelligently vary the rebuild speed based on incoming I/O traffic while maintaining availability. PowerStore uses a machine-learning algorithm and automatically adjusts the rebuild rate to prioritize host I/O when there is a drive failure to optimize performance, while maintaining reliability.

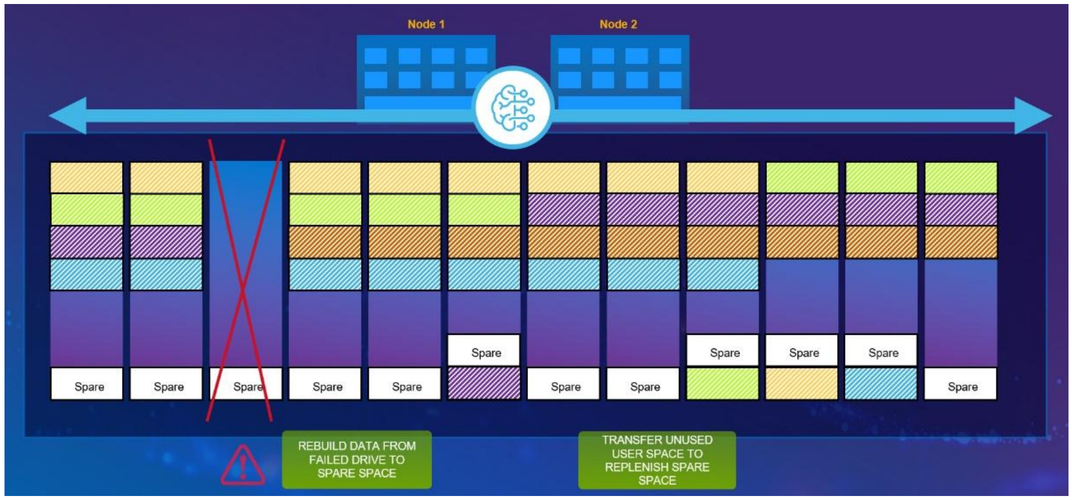

Figure 36. Rebuild data chunks to spare space after drive failure and replenish spare space with unused user space

- Flexible configurations:
  - Lower TCO with ability to expand storage by adding single drives. Capacity can be added non-disruptively, giving customers the flexibility to expand their storage by adding one or more drives based on their need
  - Flexible options to add different drive sizes based on storage need. PowerStore implements proprietary algorithms to manage drives of different sizes by optimizing the distribution of redundancy extents across multiple drives

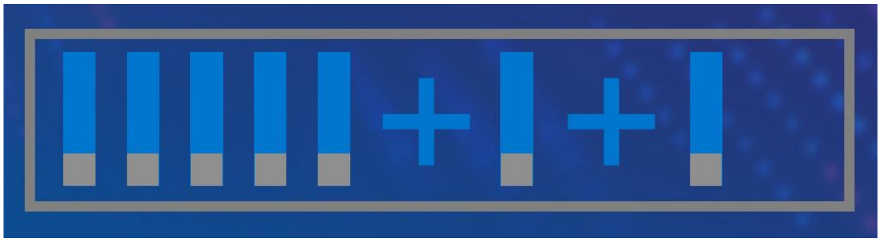

Figure 37. Single drive expansion

Block storage

The PowerStore architecture implements a fully active/active thinly provisioned mapping subsystem that enables I/Os from any path to be committed and fully consistent without the need for redirection. The primary advantage of this implementation is to reduce the time to service I/O requests if some paths were to become unavailable. This approach contrasts with competing architectures that use redirection and single node locking that can result in long-trespass times, which increases the chances of data unavailability during failover.

PowerStore implements a dual-node, shared-storage architecture that uses Asymmetric Logical Unit Access (ALUA) for SCSI-based host access and Asymmetric Namespace Access (ANA) for NVMe-oF host access. I/O requests received on any path (active/optimized or active/non-optimized) are committed locally and are fully consistent with its peer node. I/O is normally sent down an active/optimized path. However, in the rare event that all active/optimized paths become unavailable, I/O requests are processed on the local node through the active/non-optimized paths without the need to send any data to the peer node. Then, the I/O is written to the shared NVRAM write cache where the I/O is deduplicated and compressed before being written to the drives. For more information about how the I/O is written to the drives, see the document *Dell PowerStore: Data Efficiencies*.

ALUA and ANA multipathing ensures high availability. Also, the underlying fully symmetric thin provisioning architecture eliminates the complexity and overhead that is associated with making the volumes available on the surviving path which impacts time to service I/O. To use ALUA and ANA, you must install multipathing software on the host. You should configure multipathing software to use the optimized paths first and only use the non-optimized paths if there are no optimized paths available. If possible, use two separate network interface cards (NICs) or Fibre Channel host bus adapters (HBAs) on the host. This use avoids a single point of failure on the card and the server card slot.

Because the physical ports must always match on both Nodes, the same port numbers are always used for host access in the event of a failover. For example, if 4-port card port 3 on node A is currently used for host access, the same port would be used on node B in the event of a failure. Because of this, connect the same port on both nodes to the multiple switches for host multipathing and redundancy purposes.

Ethernet configuration

PowerStore includes support for NVMe/TCP, which allows users to configure Ethernet interfaces for iSCSI or NVMe/TCP host connectivity. You can deploy Ethernet interfaces in mirrored pairs to both PowerStore nodes since these interfaces do not fail over. This configuration ensures that the host has continuous access to block-level storage resources if one node becomes unavailable. In PowerStore, storage networks are configured after the cluster is created. For a robust HA environment, create additional interfaces on other ports. To remove a single point of failure at the host and switch level, verify that the PowerStore system has redundant connectivity, as shown in the following figure.

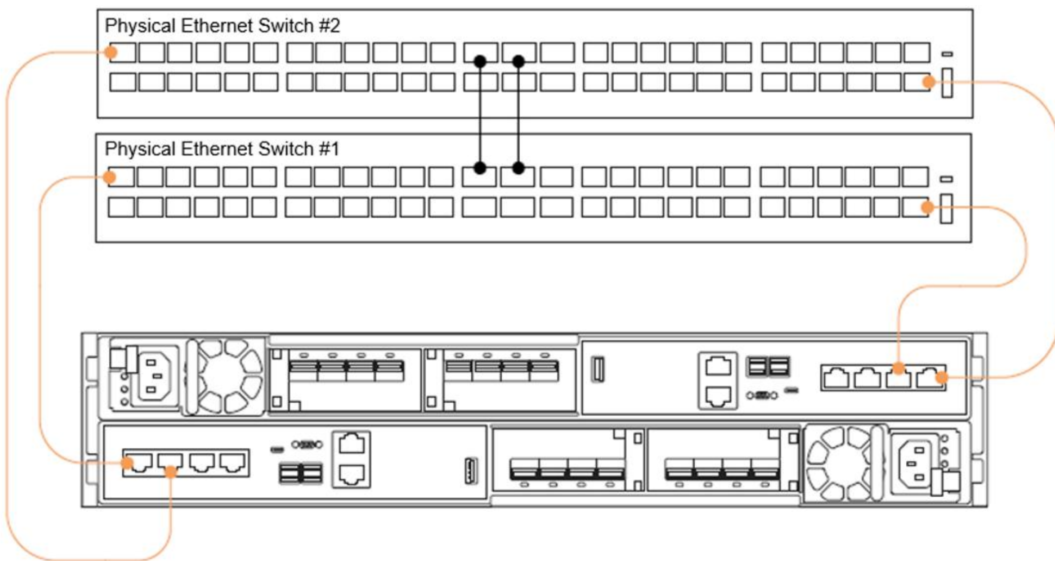

Figure 38. First two ports of the 4-port card cabled to top-of-rack switches

After completing cabling, add networks and IP addresses in PowerStore Manager from the **Settings** > **Networking** > **Network IPs** page. Figure 39 shows the first step of making sure that there are unused IP addresses configured on an existing storage network. If there are no IPs available, click the **Add** button to supply additional IP addresses. Figure 40 shows the next step for mapping extra ports for host connectivity, which can be completed on the **Hardware** > **Ports** page.

These ports obtain the next-available unused IP address from the storage network and assign them to the newly configured port. As seen in this example, newly configured ports that are assigned for host traffic are enabled in mirrored pairs. If only one port is selected, there is a notification indicating that the associated port on the peer node is also enabled for host traffic.

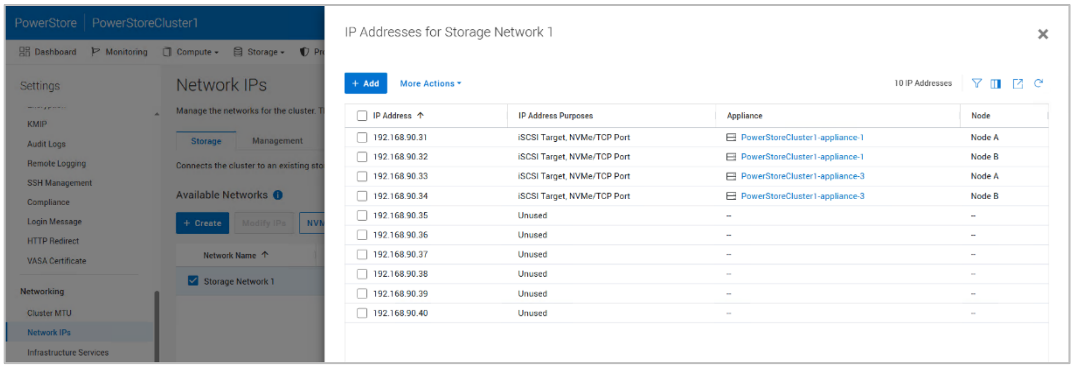

Figure 39. Verify unused storage network IP addresses

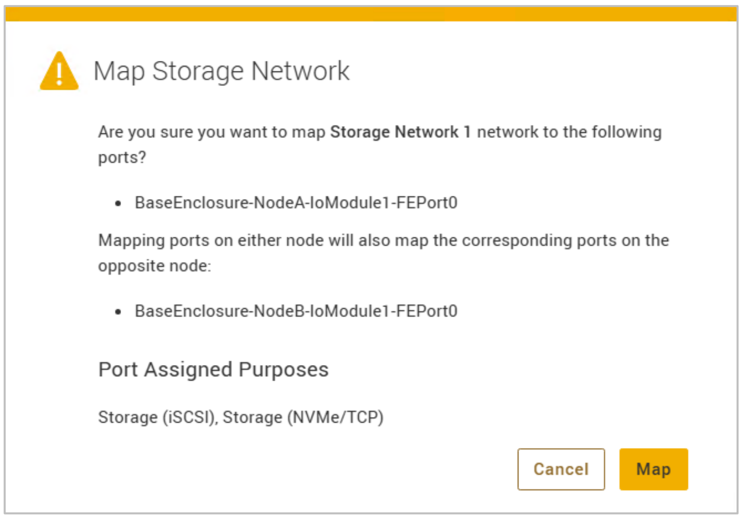

Figure 40. Map Storage Network

Fibre Channel configuration

To achieve high availability with Fibre Channel (FC) and NVMe/FC, configure at least one connection to each node. This practice enables hosts to have continuous access to block-level storage resources if one node becomes unavailable.

With FC and NVMe/FC, you must configure zoning on the switch to allow communication between the host and the PowerStore appliance. If there are multiple appliances in the cluster that are using FC, ensure that each PowerStore appliance has zones that are configured to enable host communication. Create a zone for each host HBA port to each node FC port on each appliance in the cluster. For a robust HA environment, you can zone more FC ports to provide more paths to the PowerStore appliance.

There are two locations where you can locate the port World Wide Names (WWNs) in PowerStore Manager. The first location is in the **Hardware** > **Ports** > **FE Ports Health** page, as shown in the following figure. The second location is in the **Hardware > Appliances > [Appliance] > Components** > **Rear View** page, as shown in Figure 42. For more details about configuring hosts, see the document *PowerStore Host Configuration Guide* on dell.com/powerstoredocs.

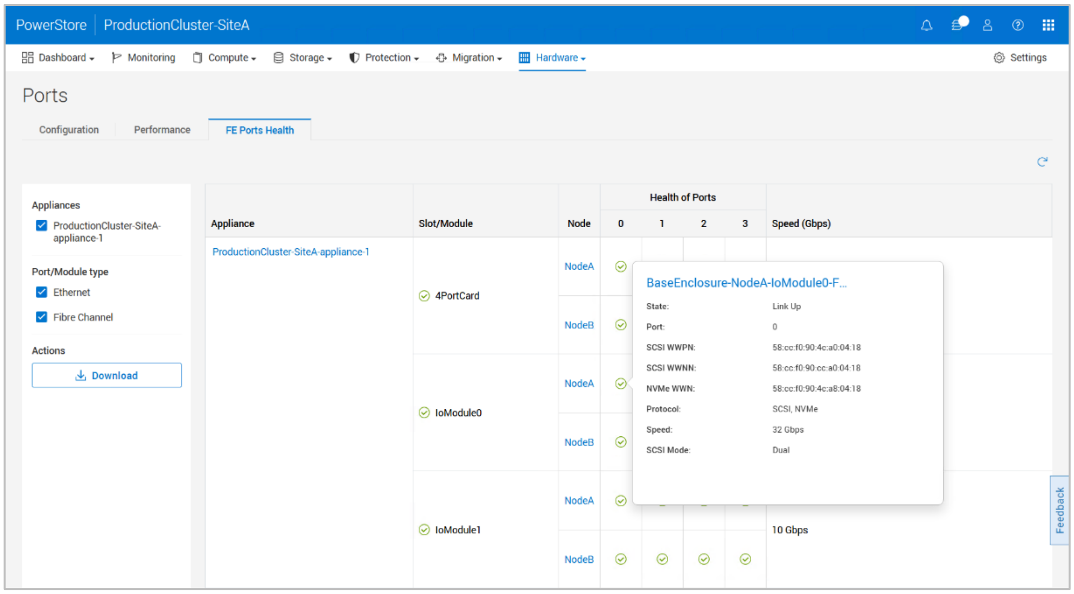

Figure 41. PowerStore Manager hardware cluster front-end ports page

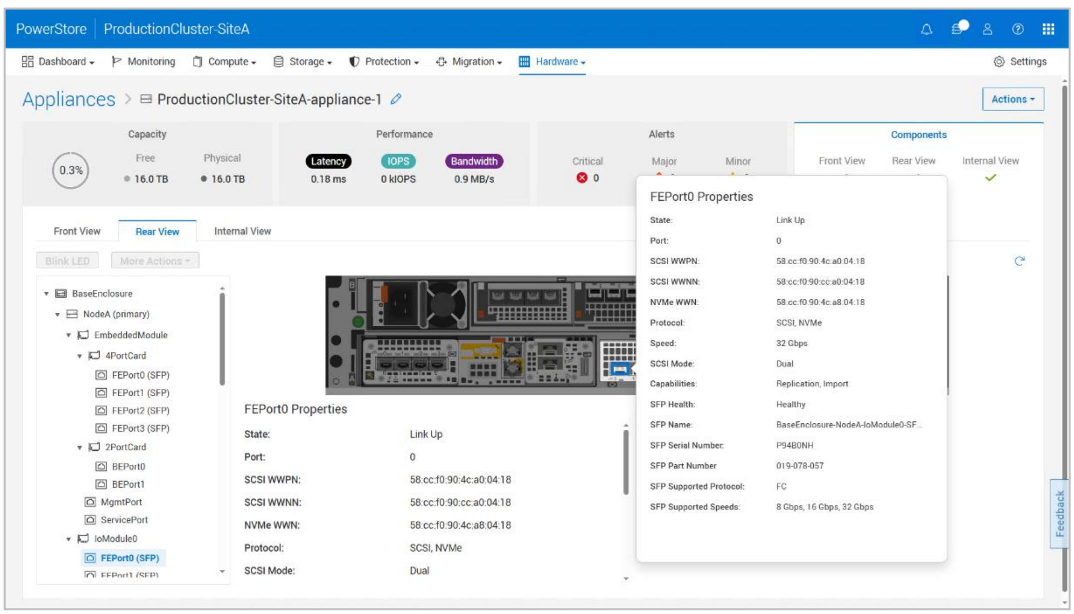

Figure 42. PowerStore Manager components rear view

Block example

When designing a highly available infrastructure, components that connect to the storage system must also be redundant. This design includes removing single points of failure at the host and switch level to avoid data unavailability due to connectivity issues. The following figure shows an example of a highly available configuration for a PowerStore system, which has no single point of failure. See the documents *PowerStore Network Planning Guide* and *PowerStore Network Configuration for PowerSwitch Series Guide* on the dell.com/powerstoredocs for detailed information about cabling and configuration of the network infrastructure.

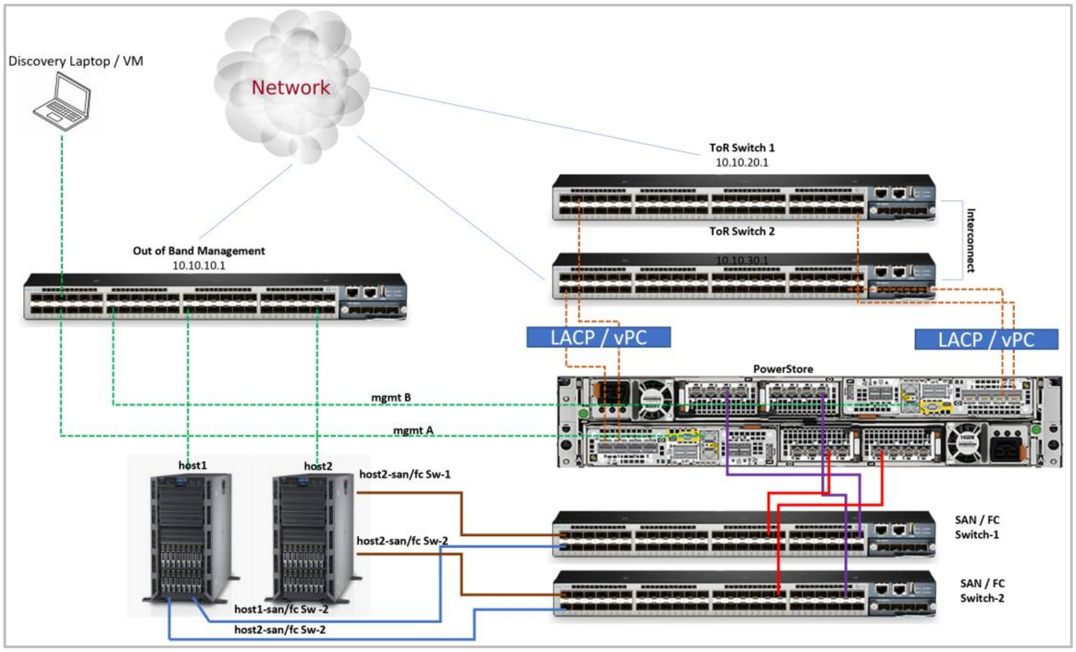

Figure 43. Highly available block configuration

File storage

PowerStore uses virtualized NAS servers to enable access to file systems, provide data separation, and act as the basis for multitenancy. The network interface of a NAS server is required to be placed on a Link Aggregation Group or a Fail-Safe Network (FSN). Link Aggregation Control Protocol (LACP) and Fail-Safe Networking (FSN) are discussed earlier in this document.

An administrator can create a NAS server that holds the configuration information for SMB, NFS, FTP, or SFTP access to the file systems. New NAS servers are automatically assigned on a round-robin basis across the available nodes. The preferred node acts as a marker to indicate the node on which the NAS server should be running, based on this algorithm. After it is provisioned, the preferred node for a NAS server never changes. The current node indicates the node on which the NAS server is running. Changing the current node moves the NAS server to a different node, which can be used for loading balancing purposes. When a NAS server is moved to a new node, all file systems on the NAS server move along with it. The following figure shows the current and preferred node columns in PowerStore Manager.

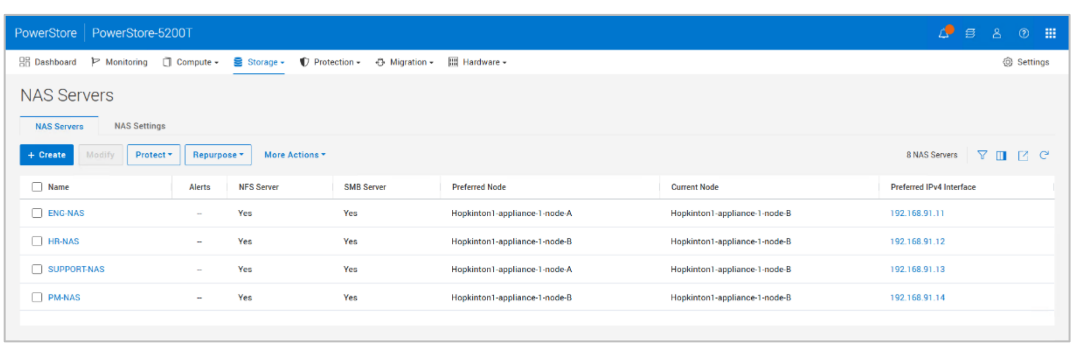

Figure 44. Current Node and Preferred Node

During a PowerStore node failure, the NAS servers automatically fail over to the surviving node. This process generally completes within 30 to 60 seconds to avoid host timeouts. After the failed node is recovered, a manual process is required to fail back the NAS servers and return to a balanced configuration.

NAS servers are automatically moved to the peer node and back during the upgrade process. After the upgrade is completed, the NAS servers return to the node that they were assigned to at the beginning of the upgrade. For more information about NAS Servers, see the document *Dell PowerStore: File Capabilities*.

VMware integration

vSphere HA

When administrators are using virtual machines on PowerStore, it is recommended to enable vSphere HA on the cluster. If a node or host in the cluster loses power, virtual machines will restart on a surviving host or node in the cluster.

vCenter connection

PowerStore appliances support integration with VMware environments by being managed and monitored from a vCenter. For PowerStore, a vCenter connection is optional after the cluster has been configured. If a vCenter connection is lost to the PowerStore cluster, management tasks through vCenter may be temporarily unavailable. Note that I/O continues to be processed for all storage objects in the PowerStore cluster.

Replication

To protect against outages at a system or data-center level, it is recommended to use replication to protect data at a remote site. This use case includes planned maintenance events, unplanned power outages, or natural disasters. PowerStore supports replication to simplify disaster recovery for block and file resources.

For block resources, including volumes, volume groups, and thin clones, asynchronous replication is included in all PowerStoreOS releases. Metro volume replication, added in PowerStoreOS 3.0, replicates data within VMware VMFS datastores spanned across two PowerStore clusters in a metro distance. With metro volume replication, concurrent active/active host I/O is possible on both sides and is replicated to the remote system. In PowerStoreOS 4.0 and later, metro support was expanded to volume groups, along with support for Windows and Linux operating systems. Synchronous replication was also added in PowerStoreOS 4.0 for volumes, volume groups, and thin clones.

For file resources, starting in PowerStoreOS 3.0, asynchronous replication can be used to protect against a storage system outage by creating a copy of data on a remote system. When an NAS server is replicated, all file systems are replicated with it. Synchronous replication, added in PowerStoreOS 4.0, is also supported on file resources.

For more details on the replication features for PowerStore, see the white papers *Dell PowerStore: Replication Technologies* and *Dell PowerStore: Metro Volume* on the Dell Technologies Info Hub.

Platform high availability

On PowerStore, each storage resource is assigned to either node A or node B for load-balancing and redundancy purposes. Besides storage-resource assignments, each node has various containers running on them that make up the PowerStore operating system. If one node becomes unavailable, its resources (storage and containers) automatically failover to the surviving node.

The time that it takes for the failover process to complete depends on several factors such as system utilization and the number of resources. The peer node assumes ownership of the resources and continues servicing I/O to avoid an extended outage. Failovers occur if the following occurs on a node:

- Node reboot: The system or a user rebooted the node.
- Hardware or software fault: The node has failed and must be replaced.
- Service mode: The system or a user placed the node into service mode. This occurs automatically when the node is unable to boot due to a hardware or software issue.
- Powered off: A user powered off the node.

Note: Manually putting a node into service mode is only available through a service script. It is not available from the PowerStore Manager, REST API, or PSTCLI.

While a node is unavailable, all the resources of the node are serviced by the peer. After the node is brought back online or the fault is corrected, block-storage resources automatically fail back to the proper node owner. File-storage resources must be failed back manually.

During a code upgrade, both nodes reboot in a coordinated manner. All resources on the rebooting node are failed over to the peer node. When the peer comes back online, the resources are failed back to their original owner. This process repeats for the second node. Users can run a pre-upgrade health check before starting a code upgrade to ensure a smooth upgrade process.

Cluster high availability

Every PowerStore appliance uses the pacemaker stack, which is used for cluster resource management and making sure that the cluster has a quorum. The pacemaker is the cluster brain; it processes and reacts to cluster events such as appliances being added or removed from the cluster, or resource events that are caused by failures. During an appliance failure, management services are still serviced if there is a quorum.

Cluster quorum

A quorum is defined as N/2+1 appliances being in active communication. If there is no quorum, management operations are temporarily lost, but data continues to be serviced if available. The following figure shows an example of a two-appliance cluster that is servicing I/O to a host with Fibre Channel connectivity but has temporarily lost management access. In this example, there is no quorum between Appliance 1 and Appliance 2. Because the appliances are servicing I/O to the host with a Fibre Channel connectivity, there is no impact on the host. However, since quorum is lost, some management services are temporarily suspended until quorum is restored:

- Access to PowerStore Manager
- Running scheduled snapshots
- Syncing replication sessions

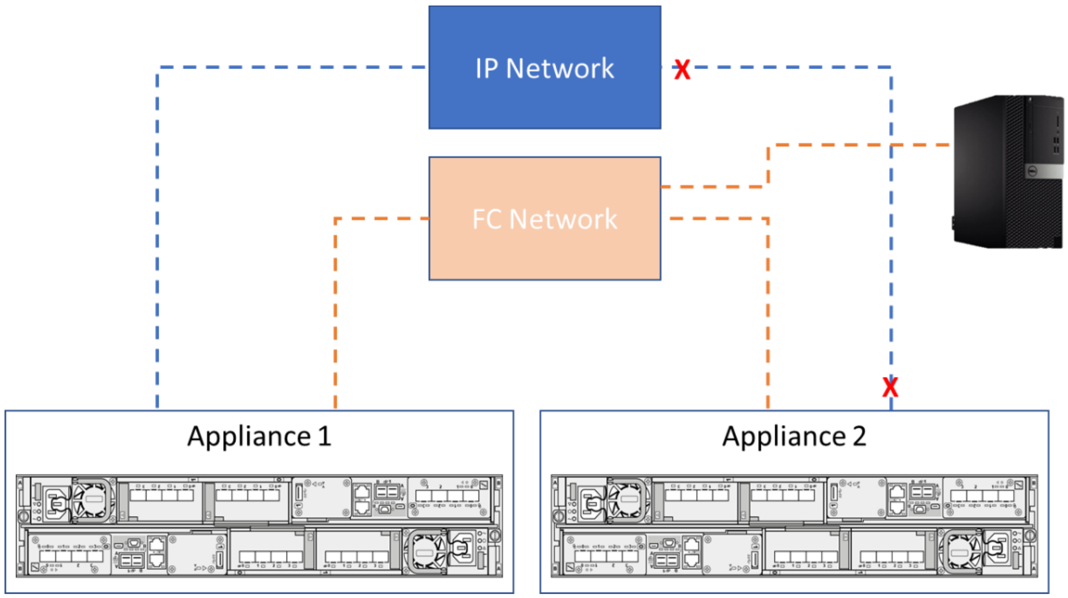

Figure 45. Two-appliance cluster with no quorum

The following figure shows a three-appliance cluster where Appliance 3 is unavailable. In this scenario, quorum is met because two appliances are still in active communication. I/O remains accessible from Appliance 1 and Appliance 2. However, since Appliance 3 is unavailable, I/O is inaccessible for that appliance.

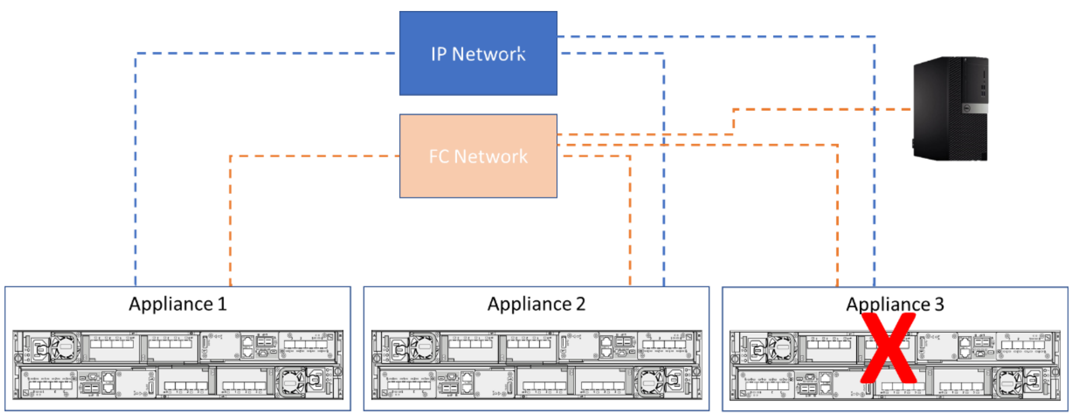

Figure 46. Three-appliance cluster with quorum

Cluster IP

The cluster IP address is a required field that is set during the initial configuration of the cluster. The cluster IP address is a highly available IP address that is used to access PowerStore Manager. The cluster IP address is typically on the primary node of the primary appliance. In the rare event that the primary appliance experiences a dual-node failure, the cluster IP address fails over to another appliance in the cluster if there is quorum. When this cluster IP address fails over, it might take several minutes to regain access to the PowerStore Manager.

Global storage IP

The Global Storage IP (GSIP) address is an optional field that can be set during or after the initial configuration of the cluster. This IP address is a global, floating storage-discovery IP. iSCSI or NVMe/TCP hosts only require one GSIP and can discover all the storage paths for all the appliances in the cluster. Otherwise, iSCSI or NVMe/TCP hosts require a list of storage IPs so that if one IP is down, the host can try the next IP.

Conclusion

Designing an infrastructure with high levels of availability ensures continuous access to business-critical data. If data becomes unavailable, day-to-day operations are impacted, which could lead to loss of productivity and revenue. PowerStore systems are designed with full redundancy across all components at both the hardware and software level. These features enable the system to be designed for 99.9999% availability. By using the clustering and high availability features in PowerStore, organizations can minimize the risk of data unavailability.

References

Dell Technologies documentation

The following Dell Technologies documentation provides other information related to this document. Access to these documents depends on your login credentials. If you do not have access to a document, contact your Dell Technologies representative.

- PowerStore Info Hub
- Dell PowerStore: Introduction to the Platform
- Dell PowerStore: Virtualization Integration
- Dell PowerStore: Replication Technologies
- Dell PowerStore: Data Efficiencies
- Dell PowerStore: File Capabilities
- Dell PowerStore: Metro Volume
- Dell.com/powerstoredocs provides detailed documentation about how to install, configure, and manage PowerStore systems, including the following:
  - PowerStore Quick Start Guide
  - PowerStore Hardware Information Guide
  - PowerStore Hardware Information Guide for 500T
  - PowerStore Host Configuration Guide
  - PowerStore Network Planning Guide
  - PowerStore Network Configuration for PowerSwitch Series Guide